BLOCH'S CONSTANT CONJECTURE IS FALSE

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Abstract. We prove that the classical Bloch constant is strictly smaller than the Ahlfors–Grunsky value, disproving famous Bloch's constant conjecture.

1. Introduction

Let $\mathcal{O}(\mathbb{D})$ be the set of holomorphic functions on the unit disk $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. For a nonconstant function $f \in \mathcal{O}(\mathbb{D})$, let $b(f)$ denote the supremum of the radii of Euclidean disks onto which f maps some subdomain of $\mathbb{D}$ conformally. Equivalently, $b(f)$ is the supremum of all $r > 0$ for which there exist $w \in \mathbb{C}$ and a holomorphic map $\psi : D(w, r) \to \mathbb{D}$ satisfying $f \circ \psi = \mathrm{id}_{D(w,r)}$, where $D(w, r) = \{\zeta \in \mathbb{C} : |\zeta - w| < r\}$. The classical Bloch constant is defined by

$$\mathcal{B} = \inf\big\{b(f) : f \in \mathcal{O}(\mathbb{D}),\ f(0) = 0,\ f'(0) = 1\big\}.$$

Bloch's theorem [6] guarantees that $\mathcal{B} > 0$. Landau [12] considered the related problem of finding disks contained in the image of a holomorphic function, which leads to the Landau constant problem.

Ahlfors and Grunsky [3] constructed a function whose inverse Riemann surface is tiled by equilateral triangles, with simple branch points lying over a triangular lattice. Their construction yields the upper bound

$$\mathcal{B} \le A_{\mathrm{AG}} := \frac{\Gamma(1/3)\Gamma(11/12)}{\sqrt{1+\sqrt{3}}\,\Gamma(1/4)} = 0.471861653452681\ldots, \tag{1.1}$$

where Γ denotes the Euler gamma function. They conjectured that equality holds in (1.1), which is known as *Bloch's constant conjecture*.

Bloch's constant conjecture. *The Bloch constant satisfies* $\mathcal{B} = A_{\mathrm{AG}}$.

Progress on lower bounds for Bloch's constant is summarized in Table 1.

Lower bound for $\mathcal{B}$	Reference
$\mathcal{B} \ge \sqrt{3}/4$	Ahlfors [1]
$\mathcal{B} > \sqrt{3}/4$	Heins [11]
$\mathcal{B} \ge \sqrt{3}/4 + 10^{-14}$	Bonk [7]
$\mathcal{B} \ge \sqrt{3}/4 + 2 \cdot 10^{-4}$	Chen–Gauthier [9]
$\mathcal{B} \ge \sqrt{3}/4 + 3 \cdot 10^{-4}$	Xiong [15]
$\mathcal{B} \ge \sqrt{3}/4 + 0.0153$	Wikström [14]

Table 1. A summary of lower bounds for Bloch's constant $\mathcal{B}$.

For the spherical metric, Bonk and Eremenko [8] proved that every nonconstant meromorphic function on $\mathbb{C}$ univalently covers spherical disks of radii arbitrarily close to $\arctan\sqrt{8}$, and that this constant is sharp. They further showed that the sharp constant is $\pi/2$ when all critical points are multiple. Unlike the classical Bloch constant problem, their covering theorem requires no derivative normalization.

We recall the classical Ahlfors–Grunsky function [3], defined by an anaytic continuation by Schwarz reflectionr of the conformal map g from an equilateral hyperbolic triangle with all three angles equal to $\pi/6$ onto a Euclidean equilateral triangle, and normalized by $g(0) = 0$ and $g'(0) = 1$. For this construction, see Yamada [16, p. 401].

Minda [13] studied more general Bloch constants, while Yanagihara [17] gave necessary conditions for extremal functions. Goodman [10] considered variations of individual branch points. Yamada [16] showed

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that any sufficiently small nonzero displacement of a simple branch point of the Ahlfors–Grunsky inverse Riemann surface strictly increases the normalized Bloch radius, provided the other branch points and the boundary remain fixed. Yanagihara [18] studied second-order variations induced by affine transformations of the full image surface. Further local minimality results were obtained by A. Baernstein II and J. Vinson [5]. In this paper, our main result disproves Bloch's constant conjecture.

Theorem 1.1. *There exist an Ahlfors–Grunsky function g, constants $\delta, c > 0$, and normalized holomorphic functions f_s on $\mathbb{D}$, $0 < s < \delta$, such that*

$$b(g) = A_{\mathrm{AG}}, \qquad b(f_s) \le A_{\mathrm{AG}}(1 - cs), \qquad f_s \longrightarrow g \quad \text{locally uniformly as } s \downarrow 0. \tag{1.2}$$

Since $\mathcal{B} \le b(f_s)$ for every $s \in (0, \delta)$, Theorem 1.1 gives the following corollary.

Corollary 1.2. *Bloch's constant satisfies $\mathcal{B} < A_{\mathrm{AG}}$.*

We also obtain an explicit upper bound for $\mathcal{B}$.

Theorem 1.3. *Bloch's constant satisfies $\mathcal{B} < A_{\mathrm{AG}} - 1.3 \times 10^{-9}$.*

Sketch of the proof. We begin with a translation surface of genus two tiled by equilateral triangles. Its developing map on the universal cover gives an Ahlfors–Grunsky function after normalization at a face center. A horizontal cylinder on the surface lifts to a strip in $\mathbb{D}$. We shear one such lift and extend the deformation to the two complementary components. Solving the resulting Beltrami equation gives holomorphic maps $\widetilde{f}_t : \mathbb{D} \to \mathbb{C}$.

An inverse-branch argument bounds the radius of every univalent disk by the largest circumradius of the deformed faces. This gives

$$b(\widetilde{f}_t) \le \frac{1}{\sqrt{3}} + O(t^2).$$

The conformal normalization has a nonzero first variation at some face center outside the sheared strip. To find such a center, we first show that the variation cannot be infinitesimally trivial on both complementary components. A boundary expansion, together with the fact that every point of $\mathbb{D}$ lies within a bounded hyperbolic distance of a face center, then gives the required center. Normalizing there and choosing the sign of t produces a decrease of first order in the derivative, while the change in the circumradius is only of second order. This yields $b(f_s) \le A_{\mathrm{AG}}(1 - cs)$ for the normalized functions f_s, and they converge locally uniformly to an Ahlfors–Grunsky function as $s \downarrow 0$.

For the explicit bound, we use a deformation of the compact surface that shears every lift of one cylinder. A Green-function comparison gives an estimate with an explicit quadratic remainder, and its first-order coefficient is expressed in terms of finitely many integrals on the compact surface. Combining the numerical estimates in Lemma 8.3 with the remainder estimate gives Proposition 8.4. The qualitative proof the main result does not use this numerical computation.

Organization of the paper. In Section 2 we collect the analytic facts used in the proof. In Sections 3 and 4 we construct the translation surface and the deformation of a single lifted cylinder. In Section 5 we study the first variation of the conformal normalization and find a face center with nonzero variation. In Section 6 we bound the radii of all univalent disks, and Section 7 we complete the proof of Theorem 1.1. In Section 8 the compact surface deformation is treated and the explicit bound is derived.

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2. Classical analytic facts and notation

We equip $\mathbb{D}$ with the Poincaré metric $2|dz|/(1 - |z|^2)$, normalized to have Gaussian curvature -1. We denote its induced distance by $d_{\mathbb{D}}$; explicitly,

$$d_{\mathbb{D}}(z, w) = 2 \operatorname{arctanh} \left| \frac{z - w}{1 - \overline{w} z} \right|, \qquad z, w \in \mathbb{D}.$$

For an open set $\Omega \subset \mathbb{C}$ and $1 \le p \le \infty$, let $L^p(\Omega)$ and $W^{1,p}(\Omega)$ denote the usual Lebesgue and Sobolev spaces of complex-valued functions, respectively. We write $W^{1,p}_{\mathrm{loc}}(\Omega)$ for the corresponding local Sobolev space, and $W^{1,2}_0(\Omega)$ for the closure of $C^\infty_c(\Omega)$ in $W^{1,2}(\Omega)$.

In the complex coordinate $z = x + iy$, let $dA = dx\,dy$ denote Euclidean area measure, and define the Wirtinger derivatives by

$$\partial_z = \frac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\bar z} = \frac{1}{2}(\partial_x + i\partial_y).$$

Derivatives of Sobolev functions are understood in the distributional sense.

An orientation-preserving homeomorphism $h : \Omega \to \Omega'$ between plane domains is called *quasiconformal* if $h \in W^{1,2}_{\mathrm{loc}}(\Omega)$ and, for some $0 \le k < 1$,

$$|h_{\bar z}| \le k|h_z| \qquad \text{almost everywhere on } \Omega.$$

A *Beltrami coefficient* on Ω is a function $\mu \in L^\infty(\Omega)$ with $\|\mu\|_\infty < 1$. The equation $h_{\bar z} = \mu h_z$ is called the *Beltrami equation* and is understood almost everywhere.

For $G \in W^{1,2}_{\mathrm{loc}}(\Omega)$, we denote its Jacobian by

$$J_G = |G_z|^2 - |G_{\bar z}|^2.$$

For a holomorphic function F, we write $\mathrm{Crit}(F) = \{z : F'(z) = 0\}$ for its critical set.

For $0 < \delta < 1$, let

$$L^\infty_\delta(\mathbb{C}) = \{\mu \in L^\infty(\mathbb{C}) : \|\mu\|_\infty < \delta\}.$$

For $1 < p < \infty$, let N_p denote the operator norm on $L^p(\mathbb{C})$ of the Beurling transform

$$Tg(z) = -\frac{1}{\pi}\,\mathrm{pv}\int_{\mathbb{C}} \frac{g(\zeta)}{(z-\zeta)^2}\,dA(\zeta).$$

Recall that $N_2 = 1$ and $N_p \to 1$ as $p \downarrow 2$.

For $R > 0$ and $p > 2$, set $\alpha_p = 1 - 2/p$ and write $D_R = D(0, R)$. Let $B_{R,p}$ be the Banach space of functions $v \in C^{\alpha_p}(\overline{D_R})$ satisfying $v(0) = 0$ and $v_{\bar z} \in L^p(D_R)$, equipped with the norm

$$\|v\|_{B_{R,p}} = \sup_{\substack{z,w \in \overline{D_R} \\ z \ne w}} \frac{|v(z) - v(w)|}{|z - w|^{\alpha_p}} + \|v_{\bar z}\|_{L^p(D_R)}.$$

We will use the following results of Ahlfors and Bers [2].

Lemma 2.1 ([2, Theorems 6 and 11])**.** (1) *For every $\mu \in L^\infty_1(\mathbb{C})$, there exists a unique quasiconformal homeomorphism $h^\mu : \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ fixing 0, 1, and ∞ and satisfying $(h^\mu)_{\bar z} = \mu (h^\mu)_z$ almost everywhere on $\mathbb{C}$. Likewise, every Beltrami coefficient on $\mathbb{D}$ determines a unique quasiconformal self-homeomorphism of $\mathbb{D}$ fixing 0 and 1.*

(2) *Let $0 < \delta < 1$ and $p > 2$ satisfy $\delta N_p < 1$. Suppose that $\Lambda \subset \mathbb{C}$ is a domain and that $\lambda \mapsto \mu_\lambda$ is holomorphic from Λ into $L^\infty_\delta(\mathbb{C})$. Then, for every $R > 0$, $\lambda \mapsto h^{\mu_\lambda}|_{\overline{D_R}}$ is holomorphic from Λ into $B_{R,p}$.*

Lemma 2.2 ([2, Lemma 1])**.** *A continuous function in $W^{1,1}_{\mathrm{loc}}(\Omega)$ satisfying $\partial_{\bar z} v = 0$ in distributions is holomorphic on Ω.*

Proposition 2.3. *Let $\Lambda \subset \mathbb{C}$ be a domain containing 0, and suppose that $\lambda \mapsto \mu_\lambda$ is holomorphic from Λ into $L^\infty_1(\mathbb{C})$, with $\mu_0 = 0$. Fix three distinct points $\zeta_1, \zeta_2, \zeta_3 \in \partial\mathbb{D}$, and let H_λ be the unique quasiconformal homeomorphism of $\widehat{\mathbb{C}}$ with Beltrami coefficient μ_λ fixing these three points.*

(1) *For every $R > 1$, there exist a neighborhood Λ_R of 0 and an exponent $p > 2$ such that*

$$\lambda \longmapsto H_\lambda|_{D_R}$$

is holomorphic from Λ_R into $W^{1,p}(D_R)$.

(2) *Suppose that $\mu_\lambda = 0$ almost everywhere on a fixed open set $\mathcal{V} \subset D_R$ for every $\lambda \in \Lambda_R$. Then $(\lambda, z) \mapsto H_\lambda(z)$ is jointly holomorphic on $\Lambda_R \times \mathcal{V}$. In particular,*

$$\partial^j_\lambda \partial^m_z H_\lambda(z) = \partial^m_z \partial^j_\lambda H_\lambda(z), \qquad j, m \ge 0,$$

and all these derivatives depend locally uniformly on (λ, z).

(3) *Suppose, in addition, that for real $t \in \Lambda_R$ the coefficients satisfy*

$$\mu_t(z) = \frac{z^2}{\bar{z}^2}\overline{\mu_t(1/\bar{z})}$$

almost everywhere on $\mathbb{C} \setminus \{0\}$. Then H_t restricts to a quasiconformal self-homeomorphism of $\mathbb{D}$.

Moreover, the variation $V = \partial_t H_t|_{t=0}$ satisfies

$$H_t = \mathrm{id} + tV + O(t^2) \quad \text{in } W^{1,p}(D_R),$$
$$\partial_{\bar{z}} V = \partial_t \mu_t|_{t=0}$$

in the distributional sense, and

$$\mathrm{Re}(\bar{\zeta} V(\zeta)) = 0, \qquad \zeta \in \partial\mathbb{D}.$$

Proof. For (1). Fix $R > 1$. Since $\mu_0 = 0$, we may choose $0 < \delta < 1$ and a neighborhood Λ_0 of 0 such that $\mu_\lambda \in L^\infty_\delta(\mathbb{C})$ for every $\lambda \in \Lambda_0$. Choose $p > 2$ sufficiently close to 2 that $\delta N_p < 1$.

Write $h_\lambda = h^{\mu_\lambda}$. By Lemma 2.1(2), $\lambda \mapsto h_\lambda$ is holomorphic with values in $B_{R',p}$ for every $R' > 0$.

We first show that, whenever $R < R'$, the restriction map

$$B_{R',p} \longrightarrow W^{1,p}(D_R)$$

is continuous. Choose $\chi \in C^\infty_c(D_{R'})$ with $\chi = 1$ on a neighborhood of $\overline{D_R}$. For $v \in B_{R',p}$, the definition of the norm gives

$$\|v\|_{L^\infty(D_{R'})} + \|v_{\bar{z}}\|_{L^p(D_{R'})} \le C\|v\|_{B_{R',p}}.$$

Applying the L^p boundedness of the Beurling transform to χv yields

$$\begin{aligned}\|(\chi v)_z\|_{L^p(\mathbb{C})} &\le N_p \|(\chi v)_{\bar{z}}\|_{L^p(\mathbb{C})} \\ &\le C\left(\|v_{\bar{z}}\|_{L^p(D_{R'})} + \|v\|_{L^\infty(D_{R'})}\right).\end{aligned}$$

Consequently,

$$\|v\|_{W^{1,p}(D_R)} \le C\|v\|_{B_{R',p}}.$$

It follows that $\lambda \mapsto h_\lambda|_{D_R}$ is holomorphic with values in $W^{1,p}(D_R)$.

Since $p > 2$, the Sobolev embedding theorem shows that $h_\lambda(\zeta_j)$ depends holomorphically on λ for $j = 1, 2, 3$. Let M_λ be the unique Möbius transformation satisfying

$$M_\lambda(h_\lambda(\zeta_j)) = \zeta_j, \qquad j = 1, 2, 3.$$

Its coefficients depend holomorphically on λ, and $M_0 = \mathrm{id}$. By uniqueness,

$$H_\lambda = M_\lambda \circ h_\lambda.$$

After shrinking Λ_0 if necessary, M_λ has no pole on a neighborhood of $h_\lambda(\overline{D_R})$. As $W^{1,p}(D_R)$ is a Banach algebra for $p > 2$, composition with this holomorphic family of Möbius transformations preserves holomorphic dependence in $W^{1,p}(D_R)$. This proves (1).

For (2). Under the hypothesis of (2), the Beltrami equation gives $(H_\lambda)_{\bar{z}} = 0$ almost everywhere on $\mathcal{V}$. By Lemma 2.2, H_λ is holomorphic on $\mathcal{V}$. Moreover, (1) and the Sobolev embedding theorem imply that $\lambda \mapsto H_\lambda(z)$ is holomorphic for each fixed $z \in \mathcal{V}$. Joint holomorphy follows from Hartogs' theorem, and the assertions concerning derivatives follow from the Cauchy integral formula.

For (3). Let $J(z) = 1/\bar{z}$. The reflection hypothesis implies that $J \circ H_t \circ J$ has the same Beltrami coefficient as H_t. Both maps fix $\zeta_1, \zeta_2, \zeta_3$, so uniqueness gives

$$J \circ H_t \circ J = H_t.$$

Thus H_t preserves $\partial\mathbb{D}$. Since $H_0 = \mathrm{id}$ and H_t depends continuously on t, we have $H_t(\mathbb{D}) = \mathbb{D}$ for all sufficiently small real t.

By (1), the family admits a Taylor expansion in $W^{1,p}(D_R)$, which gives the asserted expansion for H_t. Differentiating $(H_t)_{\bar{z}} = \mu_t (H_t)_z$ at $t = 0$, using $H_0 = \mathrm{id}$ and $\mu_0 = 0$, yields

$$\partial_{\bar{z}} V = \partial_t \mu_t|_{t=0}.$$

Finally, $|H_t(\zeta)| = 1$ for $\zeta \in \partial\mathbb{D}$. Differentiating at $t = 0$ gives

$$2\,\mathrm{Re}(\bar{\zeta} V(\zeta)) = 0,$$

as required. □

Lemma 2.4 ([4, Theorem 5.5.1])**.** *Let Ω be a plane domain, let $h : \Omega \to \Omega'$ be a quasiconformal homeomorphism, and let $G \in W^{1,2}_{\mathrm{loc}}(\Omega)$ be continuous. If $G_{\bar{z}} = \mu G_z$ and $h_{\bar{z}} = \mu h_z$ almost everywhere for the same Beltrami coefficient μ, then $G \circ h^{-1}$ is holomorphic on Ω'.*

We state the normalization of the Ahlfors–Grunsky triangle map that will be used below.

Lemma 2.5. *Let $T \subset \mathbb{D}$ be a regular hyperbolic triangle centered at 0 with angles $\pi/6$, and let Δ be a Euclidean equilateral triangle of side length 1. If $\Phi : T \to \Delta$ is a conformal map taking the vertices of T to those of Δ in cyclic order, then*

$$|\Phi'(0)| = c_0 = \frac{1}{\sqrt{3}\, A_{\mathrm{AG}}}. \tag{2.1}$$

Proof. Put $\omega = e^{2\pi i/3}$ and $\sigma = (1+\sqrt{3})^{-1/2}$. In the classical Ahlfors–Grunsky construction [3], as described in Yamada [16, p. 401], the hyperbolic triangle S with vertices $\sigma, \omega\sigma, \omega^2\sigma$ is mapped conformally onto the Euclidean triangle with vertices $A_{\mathrm{AG}}, \omega A_{\mathrm{AG}}, \omega^2 A_{\mathrm{AG}}$. The map g is normalized by $g(0) = 0$ and $g'(0) = 1$.

Since T and S are regular hyperbolic triangles with the same angles and center, they differ by a rotation of $\mathbb{D}$. The target triangle of g has circumradius A_{AG} and hence side length $\sqrt{3}\, A_{\mathrm{AG}}$. An orientation-preserving Euclidean similarity taking this triangle onto Δ therefore has derivative of modulus $1/(\sqrt{3}\, A_{\mathrm{AG}})$.

After choosing the rotation and similarity to match the prescribed vertices, their composition with g agrees with Φ. Indeed, conformal maps between two Jordan triangles are uniquely determined by the images of their three vertices. Since the source rotation has derivative of modulus 1, the chain rule gives (2.1). □

3. An equilateral translation surface

We construct the developing map from a branched double cover of a flat torus. Set

$$h = \frac{\sqrt{3}}{2}, \qquad \tau = \frac{1}{2} + ih, \qquad L = \mathbb{Z} + \tau\mathbb{Z}, \qquad L_0 = 2\mathbb{Z} + \tau\mathbb{Z}.$$

Let $E = \mathbb{C}/L_0$. Since $L_0 \cap \mathbb{R} = 2\mathbb{Z}$, the segment $[0,1]$ projects to an embedded arc in E. Take two copies of E, cut each along this arc, and identify the opposite banks of the two slits crosswise by translations. Denote the resulting compact Riemann surface by X and the natural projection by $P : X \to E$. The holomorphic differential dw on E pulls back to a holomorphic differential $\omega = P^*dw$ on X.

Lemma 3.1. *The surface X has genus two, and ω has exactly two zeros, both simple. Moreover, X admits a triangular cell decomposition with eight equilateral faces of side length 1, whose vertices are the zeros of ω.*

Let $\pi : \mathbb{D} \to X$ be a universal covering map, and let F be a primitive of $\pi^*\omega$. Then F is holomorphic on $\mathbb{D}$, and its critical points are precisely the lifts of the two zeros of ω, each of local degree two. On every closed lifted face, F is a homeomorphism onto a Euclidean equilateral triangle of side length 1.

Proof. The projection P is a connected double cover, unbranched away from the endpoints of the slit. A small loop around either endpoint interchanges the two sheets. Thus P has precisely two simple branch points, lying over 0 and 1. In a flat coordinate w on E, with $w_v \in \{0,1\}$ the relevant branch value, one may choose a local coordinate ξ on X so that

$$P(\xi) = w_v + \xi^2, \qquad \omega = 2\xi\, d\xi.$$

Hence ω has a simple zero at each branch point and has no other zeros. Since E has genus one, Riemann–Hurwitz gives

$$2g(X) - 2 = 2\big(2g(E) - 2\big) + 2 = 2,$$

so $g(X) = 2$.

We now describe the triangular faces. A fundamental parallelogram for L_0 has vertices $0, 2, 2+\tau, \tau$. Its four unit equilateral triangles have counterclockwise vertex triples

$$A_j = (j, j+1, j+\tau),$$
$$B_j = (j+1+\tau, j+\tau, j+1), \qquad j = 0, 1.$$

Indeed, $|\tau| = |1-\tau| = 1$, so every side of these triangles has length 1. The four faces and their common edges are shown in Figure 1; Figure 2 shows the opposite-side identifications that give E. The only vertices in the quotient are 0 and 1 modulo L_0.

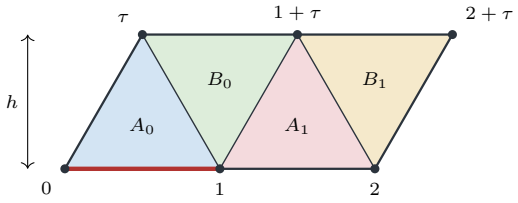


FIGURE 1. A fundamental parallelogram for L_0, divided into the four unit equilateral triangles A_0, B_0, A_1, B_1. The red edge is $[0, 1]$.

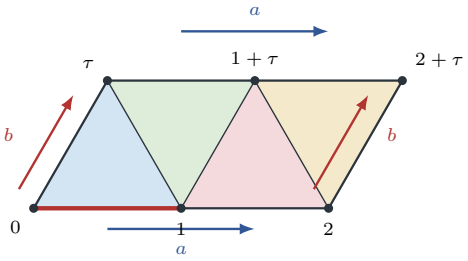


FIGURE 2. The torus $E = \mathbb{C}/L_0$ as a quotient. Opposite boundary edges bearing the same arrow label are identified by translation. The quotient has exactly two vertices, represented by 0 and 1.

The segment $[0, 1]$ is one edge of this cell decomposition. Cutting two copies of E along this edge and gluing them crosswise leaves all eight faces intact. The two identifications of the slit banks are illustrated in Figure 3. In each copy of E, six face corners meet at each vertex, giving total angle 2π. The crosswise gluing makes twelve corners meet at each of the two branch points of X, so the total flat angle there is 4π.

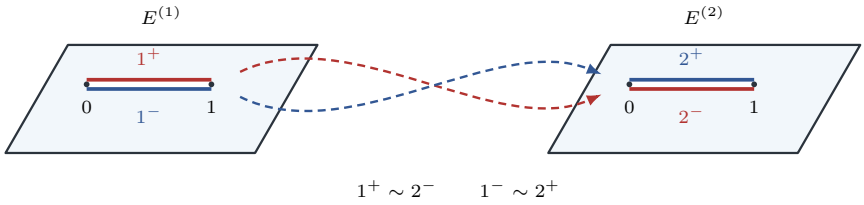


FIGURE 3. Crosswise gluing of two slit tori. For clarity, each torus is represented by a shifted fundamental parallelogram in which the slit is interior. The red and blue banks are paired as indicated.

Figure 4 shows the 4π cone angle without trying to embed a 4π Euclidean cone in the plane. It develops the angular coordinate into twelve consecutive sectors of angle $\pi/3$, which project in pairs to the six sectors around the corresponding vertex of E.

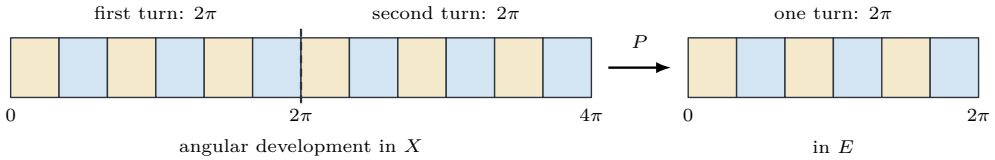


FIGURE 4. Angular development at a branch point. On X, twelve flat sectors of angle $\pi/3$ occur consecutively, for a total angle 4π. The projection P identifies sectors six apart, producing the six sectors (total angle 2π) around a vertex of E.

As $g(X) = 2$, the uniformization theorem identifies its universal cover with $\mathbb{D}$. The pullback $\pi^*\omega$ is holomorphic, and the simple connectivity of $\mathbb{D}$ gives a holomorphic primitive F. Since π is an unbranched

covering, the local form of ω above shows that, near any lift v of a branch point,

$$F(z) - F(v) = \xi(z)^2$$

for a suitable holomorphic local coordinate ξ centered at v. Thus v is a critical point of local degree two, and there are no other critical points because ω has no other zeros.

Finally, let Δ be one of the abstract Euclidean faces, with its characteristic map $\chi_\Delta : \Delta \to X$. Since Δ is simply connected, χ_Δ has a lift $\widetilde{\chi}_\Delta : \Delta \to \mathbb{D}$. On the interior of Δ,

$$d\big(F \circ \widetilde{\chi}_\Delta\big) = dw,$$

so $F \circ \widetilde{\chi}_\Delta(w) = w + c_\Delta$ for some constant $c_\Delta \in \mathbb{C}$. This identity extends to the closed face by continuity and shows that both $\widetilde{\chi}_\Delta$ and the restriction of F to its image are injective. Hence F maps every closed lifted face homeomorphically onto a unit equilateral triangle. Figure 5 illustrates the local arrangement of faces around a lifted zero: twelve faces meet at the critical point and develop onto six planar triangles, each covered twice. The lifted faces form a locally finite triangulation of $\mathbb{D}$, although images of distinct faces under F may overlap.

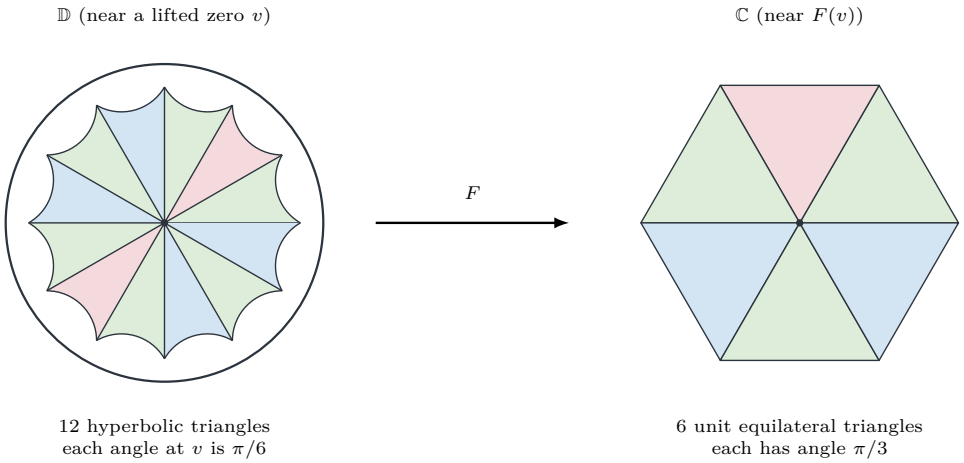


FIGURE 5. Local development around a lifted zero of ω. The left panel shows the corresponding regular hyperbolic realization of the lifted faces; twelve angles of $\pi/6$ fill 2π in the source. Their flat images give twelve angles of $\pi/3$, hence two turns around $F(v)$ and a map of local degree two. The two shaded source faces have the same planar image.

□

Let $\mathcal{P}$ be the set of inverse images, in the lifted faces, of the centers of their Euclidean images.

Proposition 3.2. *In the Poincaré metric of $\mathbb{D}$, the lifted faces are congruent regular hyperbolic triangles with angles $\pi/6$, and their hyperbolic centers are precisely the points of $\mathcal{P}$. For every $p \in \mathcal{P}$,*

$$(1 - |p|^2)|F'(p)| = c_0. \tag{3.1}$$

Moreover, there is a constant $D_0 < \infty$ such that every point of $\mathbb{D}$ lies within hyperbolic distance D_0 of a point of $\mathcal{P}$.

Proof. Let $T \subset \mathbb{D}$ be a closed regular hyperbolic triangle with angles $\pi/6$. Replace the eight abstract flat faces of X by copies of T, and glue their sides by the hyperbolic isometries with the same endpoint identifications as in X. Denote the resulting surface by Y. The metric is smooth across each open edge, as is seen by reflection in the geodesic side. By Lemma 3.1, each vertex of X is a simple zero of ω and hence has total flat angle 4π. Thus twelve face corners meet at each vertex. The corresponding angle in Y is $12(\pi/6) = 2\pi$, so there is no cone singularity. Hence Y is a smooth compact hyperbolic surface with the same underlying topology as X.

On each face choose the conformal map to the corresponding flat triangle that matches the vertices in cyclic order. The boundary extension theorem for Jordan domains gives a homeomorphism of the closed triangles, and the three vertex images determine the map uniquely. Conjugating by corresponding rotations, or by a pair of corresponding median reflections, gives another conformal map with the same vertex images. Uniqueness therefore shows that the face maps intertwine these symmetries.

Let ℓ be the hyperbolic side length. When each side is parametrized by arclength in the positive boundary direction, its boundary map is described by the same increasing function $\rho : [0, \ell] \to [0, 1]$, independently of the face and side. The median reflection interchanging the endpoints of a side gives

$$\rho(0) = 0, \qquad \rho(\ell) = 1, \qquad \rho(\ell - s) = 1 - \rho(s) \quad (0 \le s \le \ell).$$

The side identifications reverse the boundary directions: they pair the parameters s and $\ell - s$ on the hyperbolic sides, and the parameters r and $1 - r$ on the flat sides. The displayed identity therefore shows that the face maps agree on every paired side. They consequently define a homeomorphism $\Psi : Y \to X$.

At an interior point of an edge, choose a flat coordinate on X in which the edge is a straight segment. Schwarz reflection across this segment and the corresponding geodesic in Y extends each of the two face maps holomorphically across the edge. The extensions agree by the identity theorem, since their values agree on the edge. Thus Ψ is holomorphic away from the vertices. If y is a vertex, choose an ordinary holomorphic coordinate ξ on X centered at $\Psi(y)$. The function $\xi \circ \Psi$ is holomorphic on a punctured neighborhood of y and tends to 0 at y. The removable singularity theorem extends it holomorphically across y. Hence Ψ is a holomorphic homeomorphism, and therefore a conformal equivalence.

By uniqueness of the metric of curvature -1 in a given conformal class on a compact surface of genus two, Ψ identifies the hyperbolic metric on Y with the Poincaré metric on X. The universal covering $\pi : \mathbb{D} \to X$ is a local isometry for the Poincaré metrics, so every lifted face is an isometric copy of T. The equivariance under rotations through $2\pi/3$ shows that F maps the hyperbolic center of each lifted face to the center of its Euclidean image. This identifies the hyperbolic face centers with $\mathcal{P}$.

Fix $p \in \mathcal{P}$, let Q be its lifted face, and choose $\phi \in \mathrm{Aut}(\mathbb{D})$ with $\phi(0) = p$. Then $\phi^{-1}(Q)$ is a regular hyperbolic triangle centered at 0, and $F \circ \phi$ maps its interior conformally onto a unit equilateral triangle, with the vertices in cyclic order. Lemma 2.5 and the chain rule give

$$c_0 = |(F \circ \phi)'(0)| = |F'(p)|\,|\phi'(0)| = (1 - |p|^2)|F'(p)|,$$

which proves (3.1).

Finally, since T is compact, its hyperbolic diameter $D_0 := \mathrm{diam}_{\mathbb{D}}(T)$ is finite. Given $z \in \mathbb{D}$, choose a lifted closed face Q containing z, and let $p_Q \in \mathcal{P}$ be its center. Congruence of the lifted faces gives

$$d_{\mathbb{D}}(z, p_Q) \le \mathrm{diam}_{\mathbb{D}}(Q) = D_0.$$

□

We next determine the Bloch radius of the developing map F. Let $q : \mathbb{C} \to E$ be the quotient map. Choose a lift $v_0 \in \mathbb{D}$ of the branch point of P over $0 \in E$, and normalize F by $F(v_0) = 0$. Since $dF = \pi^*\omega$ and $\omega = P^*dw$, the maps $q \circ F$ and $P \circ \pi$ have the same differential and agree at v_0. Hence

$$q \circ F = P \circ \pi. \tag{3.2}$$

Lemma 3.3. *The restriction $F : \mathbb{D} \setminus \mathrm{Crit}(F) \to \mathbb{C} \setminus L$ is a holomorphic covering map. Moreover, $b(F) = r_0 = 1/\sqrt{3}$.*

Proof. Since $L = L_0 \cup (1 + L_0)$, we have $q^{-1}(\{0, 1\}) = L$. By (3.2) and Lemma 3.1,

$$F^{-1}(L) = \pi^{-1}\big(P^{-1}(\{0, 1\})\big) = \mathrm{Crit}(F). \tag{3.3}$$

In particular, F maps $\mathbb{D} \setminus \mathrm{Crit}(F)$ into $\mathbb{C} \setminus L$ and is locally biholomorphic there.

We show that this restriction is a covering map. Set $E^\circ = E \setminus \{0, 1\}$. Both $q : \mathbb{C} \setminus L \to E^\circ$ and $P \circ \pi : \mathbb{D} \setminus \mathrm{Crit}(F) \to E^\circ$ are covering maps. The latter follows by composing the unbranched double cover $P : X \setminus P^{-1}(\{0, 1\}) \to E^\circ$ with the restriction of π.

First, F is surjective onto $\mathbb{C} \setminus L$. Fix $z_0 \in \mathbb{D} \setminus \mathrm{Crit}(F)$ and let γ be any path in $\mathbb{C} \setminus L$ starting at $F(z_0)$. Lift $q \circ \gamma$ through $P \circ \pi$, starting at z_0, and denote the lift by $\widetilde{\gamma}$. By (3.2), $F \circ \widetilde{\gamma}$ and γ are lifts of the same path under q with the same initial point. Uniqueness of path lifting gives

$$F \circ \widetilde{\gamma} = \gamma.$$

Since $\mathbb{C} \setminus L$ is path connected, this proves surjectivity.

Now fix $w \in \mathbb{C} \setminus L$. Choose a simply connected neighborhood W of $q(w)$ that is evenly covered by both q and $P \circ \pi$, and let V be the component of $q^{-1}(W)$ containing w.

If U is a component of $(P \circ \pi)^{-1}(W)$, then $F(U)$ lies in a single component V_U of $q^{-1}(W)$. Moreover,

$$F|_U = (q|_{V_U})^{-1} \circ (P \circ \pi)|_U,$$

so F maps U biholomorphically onto V_U. Consequently, $F^{-1}(V)$ is the disjoint union of those components U for which $V_U = V$, each mapped biholomorphically onto V. Thus V is evenly covered by F, proving the first assertion.

It remains to compute $b(F)$. The Voronoi cells of the triangular lattice L are regular hexagons with circumradius $r_0 = 1/\sqrt{3}$. Their vertices are the centers of the unit equilateral triangles in the lattice tiling. Hence

$$\sup_{w\in\mathbb{C}} \operatorname{dist}(w, L) = r_0.$$

In particular, if w_0 is the center of a unit equilateral triangle, then $D(w_0, r_0)$ is disjoint from L. Since this disk is simply connected and F is a covering map over $\mathbb{C} \setminus L$, there exists a holomorphic inverse branch of F on $D(w_0, r_0)$. Therefore $b(F) \geq r_0$.

Conversely, let $r > r_0$ and suppose that there is a holomorphic map $\psi : D(w, r) \to \mathbb{D}$ satisfying $F \circ \psi = \mathrm{id}$. By the covering-radius property of L, there exists $\ell \in L \cap D(w, r)$. Equation (3.3) implies that $\psi(\ell) \in \mathrm{Crit}(F)$. Differentiating $F \circ \psi = \mathrm{id}$ at ℓ, we obtain

$$1 = F'(\psi(\ell))\psi'(\ell) = 0,$$

a contradiction. Thus $b(F) \leq r_0$, and the proof is complete. □

For each $p \in \mathcal{P}$, let $\varphi_p(z) = (z + p)/(1 + \bar{p}z)$ and define

$$g_p(z) = \frac{F(\varphi_p(z)) - F(p)}{(1 - |p|^2)F'(p)}.$$

Then $g_p(0) = 0$ and $g_p'(0) = 1$. By Proposition 3.2, the preimage under φ_p of the face containing p is a regular hyperbolic triangle centered at 0 with angles $\pi/6$. The function g_p maps this triangle conformally onto a Euclidean equilateral triangle centered at the origin. By Lemma 2.5 and (3.1), the latter has circumradius A_{AG}. Uniqueness of the conformal triangle map, followed by Schwarz reflection across its sides, identifies g_p with an Ahlfors–Grunsky function, up to a simultaneous rotation of the source and target.

Since precomposition with a disk automorphism preserves the Bloch radius, while multiplication by a constant scales it by its modulus, Lemma 3.3 and (3.1) give

$$b(g_p) = \frac{b(F)}{(1 - |p|^2)|F'(p)|} = \frac{r_0}{c_0} = A_{\mathrm{AG}}.$$

Thus every face center determines a normalized Ahlfors–Grunsky function with Bloch radius A_{AG}. This conclusion uses only the explicit construction and does not depend on Bloch's constant conjecture.

4. A lifted cylinder and its shear

The first copy of E contains the open horizontal cylinder

$$C = \{x + iy \pmod{2\mathbb{Z}} : 0 < y < h\},$$

viewed as a subset of X. It contains no zero of ω, and the integral of ω along an oriented core curve is 2. Fix a component U of $\pi^{-1}(C)$.

Proposition 4.1. *After subtracting a suitable element of L_0 from F, the restriction $F|_U$ is a conformal map onto*

$$S_h = \{w \in \mathbb{C} : 0 < \operatorname{Im} w < h\}.$$

Its inverse $k : S_h \to U$ extends to a proper embedding

$$\bar{k} : \overline{S_h} \longrightarrow \mathbb{D}, \qquad \overline{S_h} = \{w \in \mathbb{C} : 0 \leq \operatorname{Im} w \leq h\}.$$

The boundary curves

$$\Gamma_0 = \bar{k}(\mathbb{R}), \qquad \Gamma_h = \bar{k}(\mathbb{R} + ih)$$

are disjoint in $\mathbb{D}$ and have the same two distinct ideal endpoints $a, b \in \partial\mathbb{D}$.

Moreover, $\mathbb{D} \setminus (\Gamma_0 \cup \Gamma_h)$ has exactly three connected components: U and two domains Ω_0, Ω_h, where Γ_0 separates Ω_0 from U, and Γ_h separates Ω_h from U.

Proof. For orientation, see Figure 6.

The strip coordinate. The fundamental group of C is infinite cyclic. A loop winding n times around its core has ω-period $2n$. Since ω is closed, every null-homotopic loop in X has period zero. Thus the inclusion $C \hookrightarrow X$ induces an injection on fundamental groups. It follows that

$$\pi|_U : U \to C$$

is the universal covering of C.

The standard universal covering of C is the quotient map $S_h \to C$ given by the flat coordinate modulo horizontal translation by 2. Consequently, there is a conformal isomorphism

$$k : S_h \longrightarrow U$$

such that $\pi(k(w))$ is represented by w in the first copy of E. Since $\omega = dw$ on C, we have

$$d(F \circ k) = dw.$$

Hence

$$F(k(w)) = w + \ell$$

for some constant $\ell \in \mathbb{C}$. Let $q : \mathbb{C} \to E$ be the quotient map. The identity

$$q \circ F = P \circ \pi$$

implies that

$$q(F(k(w))) = q(w),$$

and therefore $\ell \in L_0$. Replacing F by $F - \ell$ preserves the identity $q \circ F = P \circ \pi$ and gives

$$F(k(w)) = w$$

throughout S_h. In particular, $F|_U$ is conformal onto S_h.

The translation $w \mapsto w + 2$ is a deck transformation of $S_h \to C$. It therefore corresponds to a nontrivial deck transformation γ of π satisfying

$$k(w + 2) = \gamma k(w), \qquad F \circ \gamma = F + 2. \tag{4.1}$$

The second identity holds on U by the first, and then on all of $\mathbb{D}$ by the identity theorem.

Extension to the boundary. Let

$$Q = [0, 2] + i[0, h].$$

The map from the interior of Q into X, obtained from the first copy of E, extends continuously to Q by taking limits from within the cylinder. Along the slit, these limits lie on the corresponding bank of the cut torus, while at the endpoints of the slit they converge to the branch points of P. Thus the extension is well defined and continuous on all of Q.

Since Q is simply connected and

$$\pi : \mathbb{D} \to X$$

is an ordinary covering map, this extension lifts continuously to $\mathbb{D}$. Choosing the lift to agree with k on the interior gives a continuous map

$$\bar{k} : Q \to \mathbb{D}.$$

By continuity, the identity $F(k(w)) = w$ extends to

$$F(\bar{k}(w)) = w, \qquad w \in Q.$$

In particular, $\bar{k}$ is injective on Q.

The first identity in (4.1) extends to the vertical sides of Q by continuity. Hence we may extend $\bar{k}$ to the entire closed strip by

$$\bar{k}(w + 2n) = \gamma^n \bar{k}(w), \qquad w \in Q, \quad n \in \mathbb{Z}.$$

These definitions agree on adjacent rectangles, so the extension is continuous. Moreover,

$$F(\bar{k}(w)) = w$$

holds throughout $\overline{S_h}$, which proves injectivity. In particular, the images of the two boundary lines are disjoint.

Ideal endpoints and separation. Let

$$K = \bar{k}(Q).$$

Then K is compact in $\mathbb{D}$, and

$$\bar{k}(\overline{S_h}) = \bigcup_{n \in \mathbb{Z}} \gamma^n K.$$

Since the deck group acts properly discontinuously on $\mathbb{D}$, the family $\{\gamma^n K : n \in \mathbb{Z}\}$ is locally finite. It follows that $\bar{k}$ is proper. Being continuous, injective, and proper, $\bar{k}$ is a topological embedding.

The compact surface X has genus two, so its universal covering group is a cocompact torsion-free Fuchsian group. Every nontrivial element of such a group is hyperbolic. In particular, γ has two distinct fixed points $a, b \in \partial\mathbb{D}$, which we label so that

$$\gamma^n z \longrightarrow b \quad (n \to +\infty), \qquad \gamma^n z \longrightarrow a \quad (n \to -\infty).$$

The convergence is uniform for z in compact subsets of $\mathbb{D}$, hence uniform on K. Together with the equivariance of $\bar{k}$, this shows that both Γ_0 and Γ_h tend to a and b at their respective ends.

Properness also shows that the image of the closed strip is closed relative to $\mathbb{D}$. Since every point on its boundary lines is a limit of points of U, we obtain

$$\overline{U} \cap \mathbb{D} = \bigcup_{n\in\mathbb{Z}} \gamma^n K, \qquad \overline{U} \cap \partial\mathbb{D} = \{a, b\}. \tag{4.2}$$

Each of Γ_0 and Γ_h is therefore a simple proper arc in $\mathbb{D}$ whose closure in $\overline{\mathbb{D}}$ is a crosscut joining a to b. The two crosscuts have disjoint interiors. By the Jordan separation theorem, their union divides $\mathbb{D}$ into three connected components.

The component between the two curves is U, because $\bar{k}$ is an embedding of the closed strip and

$$\partial U \cap \mathbb{D} = \Gamma_0 \cup \Gamma_h.$$

The remaining components lie on the opposite sides of Γ_0 and Γ_h from U; denote them by Ω_0 and Ω_h, respectively. This proves the proposition. □

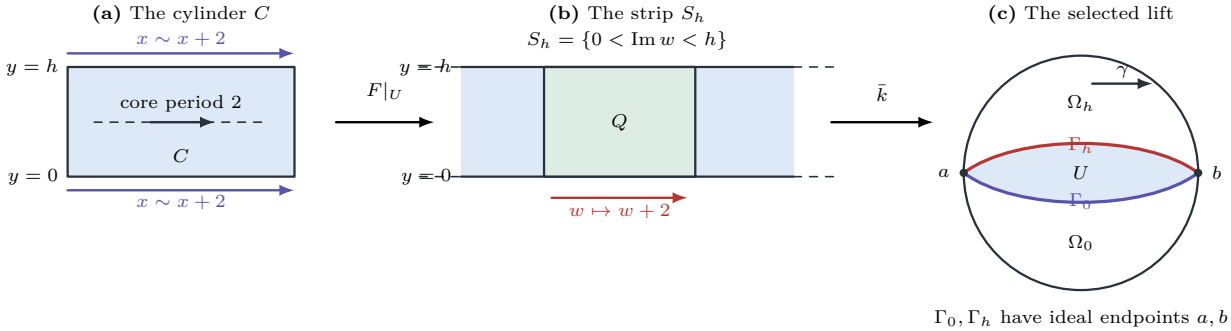


FIGURE 6. The cylinder C, the strip coordinate, and the selected lift $U \subset \mathbb{D}$. In panel (a), the vertical sides are identified by translation by 2. In panel (b), the rectangle $Q = [0,2] + i[0,h]$ is a fundamental domain for the translation $w \mapsto w+2$. In panel (c), the boundary curves Γ_0 and Γ_h have the same ideal endpoints a, b and separate $\mathbb{D}$ into the three regions U, Ω_0, Ω_h. The picture is schematic: the corners at lifted branch points are suppressed.

At a boundary vertex, the selected strip occupies exactly one flat sector of angle π. Since the local degree of F there is two, this corresponds to an angle $\pi/2$ in a regular local coordinate on $\mathbb{D}$. Thus each boundary curve has a corner at such a vertex. In particular, $\Gamma_0 \cup \Gamma_h$ is not the union of all branches of the level set of $\operatorname{Im} F$. Other lifts of horizontal cylinders may lie in Ω_0 or Ω_h and will not be sheared. Figure 6 summarizes the geometry of Proposition 4.1.

Define a continuous real-valued function on $\mathbb{D}$ by

$$u = 0 \text{ on } \Omega_0, \qquad u = \operatorname{Im} F \text{ on } U, \qquad u = h \text{ on } \Omega_h, \tag{4.3}$$

with the common limiting values on the boundary curves. For real t, set

$$G_t = F + tu.$$

Lemma 4.2. *The function u belongs to $W^{1,\infty}_{\mathrm{loc}}(\mathbb{D})$ and satisfies $du = \mathbf{1}_U \, d\operatorname{Im} F$ almost everywhere on $\mathbb{D}$. For every $t \in \mathbb{R}$, the map $G_t = F + tu$ is quasiregular, with Beltrami coefficient*

$$\mu_t = \frac{it}{2-it}\,\mathbf{1}_U \frac{\overline{F'}}{F'}, \qquad \|\mu_t\|_\infty = \frac{|t|}{\sqrt{4+t^2}}, \qquad J_{G_t} = |F'|^2. \tag{4.4}$$

The quotient $\overline{F'}/F'$ is taken to be zero at the zeros of F'.

Proof. By Proposition 4.1, the boundary curves of U consist of edges of the lifted triangulation, possibly meeting at vertices. Since the triangulation is locally finite, every compact subset of $\mathbb{D}$ meets only finitely many such edges.

The function u is smooth on each of U, Ω_0, and Ω_h, and its traces agree across their common boundaries. Indeed, $\operatorname{Im} F$ tends to 0 on Γ_0 and to h on Γ_h. Integration by parts on the individual faces therefore shows

that the boundary terms cancel, so the distributional derivatives of u agree with its piecewise classical derivatives. In particular,

$$du = \mathbf{1}_U \, d\operatorname{Im} F$$

almost everywhere on $\mathbb{D}$.

These derivatives are locally bounded. To see this near a vertex v, recall that F has local degree two there, so $F'(z) = O(|z - v|)$ as $z \to v$. Hence the piecewise gradients of u remain bounded near v. Away from the vertices, local boundedness follows directly from the holomorphy of F. Thus $u \in W^{1,\infty}_{\mathrm{loc}}(\mathbb{D})$.

Since F is holomorphic, we have

$$\partial_z \operatorname{Im} F = \frac{F'}{2i}, \qquad \partial_{\bar{z}} \operatorname{Im} F = -\frac{\overline{F'}}{2i}.$$

Consequently,

$$(G_t)_z = \left(1 - \frac{it}{2}\mathbf{1}_U\right) F', \qquad (G_t)_{\bar{z}} = \frac{it}{2}\mathbf{1}_U \overline{F'}$$

almost everywhere on $\mathbb{D}$. Where $F' \neq 0$, taking the quotient gives

$$\frac{(G_t)_{\bar{z}}}{(G_t)_z} = \frac{it}{2 - it}\,\mathbf{1}_U \frac{\overline{F'}}{F'}.$$

Since the zeros of F' are discrete, this identity determines the Beltrami coefficient almost everywhere. Moreover, U is nonempty and $|\overline{F'}/F'| = 1$ wherever $F' \neq 0$, so

$$\|\mu_t\|_\infty = \left|\frac{it}{2 - it}\right| = \frac{|t|}{\sqrt{4 + t^2}} < 1.$$

Finally, the expressions for the Wirtinger derivatives give

$$\begin{aligned} J_{G_t} &= |(G_t)_z|^2 - |(G_t)_{\bar{z}}|^2 \\ &= \left(\left|1 - \frac{it}{2}\mathbf{1}_U\right|^2 - \frac{t^2}{4}\mathbf{1}_U\right)|F'|^2 \\ &= |F'|^2, \end{aligned}$$

where we used that t is real and $\mathbf{1}_U^2 = \mathbf{1}_U$.

Thus G_t is a nonconstant $W^{1,2}_{\mathrm{loc}}$ map satisfying $|(G_t)_{\bar{z}}| \le \|\mu_t\|_\infty |(G_t)_z|$ almost everywhere, with $\|\mu_t\|_\infty < 1$. Hence G_t is quasiregular. □

5. Conformal normalization and its first variation

We solve the Beltrami equation associated with the shear and study the first variation of the hyperbolic derivative on the complementary domains.

5.1. **The parameter family.** Set $\nu = \mathbf{1}_U \overline{F'}/F'$ on $\mathbb{D}$, with the quotient defined to be zero at the zeros of F'. For $|z| > 1$, define its reflection by $\nu^{\#}(z) = (z^2/\bar{z}^2)\overline{\nu(1/\bar{z})}$. For $|\lambda| < 1/2$, consider the Beltrami coefficient

$$M_\lambda(z) = \begin{cases} \dfrac{i\lambda}{2 - i\lambda}\nu(z), & |z| < 1, \\ -\dfrac{i\lambda}{2 + i\lambda}\nu^{\#}(z), & |z| > 1. \end{cases} \tag{5.1}$$

Its values on $\partial\mathbb{D}$ are immaterial. Since $\|\nu\|_\infty = \|\nu^{\#}\|_\infty = 1$, we have

$$\|M_\lambda\|_\infty \le \frac{|\lambda|}{2 - |\lambda|} < \frac{1}{3}.$$

Moreover, $\lambda \mapsto M_\lambda$ is holomorphic as an $L^\infty(\mathbb{C})$-valued map.

Fix three distinct points $\zeta_1, \zeta_2, \zeta_3 \in \partial\mathbb{D} \setminus \{a, b\}$. Let H_λ be the unique quasiconformal homeomorphism of $\widehat{\mathbb{C}}$ with Beltrami coefficient M_λ fixing these three points. For real t, we write $H_t = H_\lambda|_{\lambda = t}$.

Proposition 5.1. *For every real $|t| < 1/2$, the map H_t restricts to a quasiconformal self-homeomorphism of $\mathbb{D}$, and $\widetilde{f}_t = G_t \circ H_t^{-1}$ is holomorphic on $\mathbb{D}$.*

Moreover, there exists $p_* > 2$ such that $V = \partial_t H_t|_{t=0}$ belongs to $W^{1,p_*}_{\mathrm{loc}}(\mathbb{D})$ and satisfies

$$H_t = \mathrm{id} + tV + O(t^2) \quad \text{locally in } W^{1,p_*}, \qquad \partial_{\bar{z}} V = \frac{i}{2}\mathbf{1}_U \frac{\overline{F'}}{F'}. \tag{5.2}$$

Both H_t and V are holomorphic on $\Omega_0 \cup \Omega_h$ and extend holomorphically across every closed subarc of $\partial\mathbb{D} \setminus \{a, b\}$. Furthermore,

$$\operatorname{Re}(\bar{\zeta} V(\zeta)) = 0, \qquad \zeta \in \partial\mathbb{D} \setminus \{a, b\}. \tag{5.3}$$

Proof. We first show that H_t preserves the unit disk. Let $J(z) = 1/\bar{z}$ denote reflection in $\partial\mathbb{D}$. For real t, the definition (5.1) gives

$$M_t(z) = \frac{z^2}{\bar{z}^2} \overline{M_t(1/\bar{z})}$$

almost everywhere on $\mathbb{C} \setminus \{0\}$. Consequently, $J \circ H_t \circ J$ has the same Beltrami coefficient as H_t. Since J fixes each ζ_j, both maps have the same normalization. Uniqueness in Lemma 2.1(1) gives

$$H_t(1/\bar{z}) = \frac{1}{\overline{H_t(z)}}.$$

In particular, H_t preserves $\partial\mathbb{D}$. As $H_0 = \mathrm{id}$ and H_t depends continuously on t, it follows that $H_t(\mathbb{D}) = \mathbb{D}$.

On $\mathbb{D}$, the coefficient M_t agrees with μ_t in (4.4). Thus Lemma 4.2 gives

$$(G_t)_{\bar{z}} = \mu_t (G_t)_z, \qquad (H_t)_{\bar{z}} = \mu_t (H_t)_z$$

almost everywhere. Since G_t is continuous and belongs to $W^{1,2}_{\mathrm{loc}}(\mathbb{D})$, Lemma 2.4 implies that $\widetilde{f}_t = G_t \circ H_t^{-1}$ is holomorphic on $\mathbb{D}$.

We next establish the variation formula. The family (5.1) satisfies $M_0 = 0$, and hence $H_0 = \mathrm{id}$. By Proposition 2.3(1), there is $p_* > 2$ such that $\lambda \mapsto H_\lambda$ is holomorphic near 0 with values in W^{1,p_*} on every fixed bounded disk. Its Taylor expansion therefore gives

$$H_t = \mathrm{id} + tV + O(t^2)$$

locally in W^{1,p_*}. Differentiating the Beltrami equation

$$(H_t)_{\bar{z}} = M_t (H_t)_z$$

at $t = 0$, using $H_0 = \mathrm{id}$ and $M_0 = 0$, we obtain

$$\partial_{\bar{z}} V = \partial_t M_t|_{t=0} = \frac{i}{2} \mathbf{1}_U \frac{\overline{F'}}{F'}$$

in the distributional sense. This proves (5.2).

Since $M_\lambda = 0$ on $\Omega_0 \cup \Omega_h$, Proposition 2.3(2) shows that H_t and V are holomorphic on these domains. To obtain the boundary extension, recall from (4.2) that $\overline{U} \cap \partial\mathbb{D} = \{a, b\}$. Thus every closed subarc $J_0 \subset \partial\mathbb{D} \setminus \{a, b\}$ has a neighborhood in $\widehat{\mathbb{C}}$ disjoint from both $\overline{U}$ and its reflection. The coefficients M_λ vanish identically on this neighborhood for all $|\lambda| < 1/2$.

By Proposition 2.3(2), the normalized solutions and their parameter derivatives are holomorphic across J_0. In particular, V extends holomorphically across every such arc.

Finally, H_t preserves the unit circle, so $|H_t(\zeta)|^2 = 1$ for $\zeta \in \partial\mathbb{D}$. Differentiating at $t = 0$ gives

$$2 \operatorname{Re}\big(\overline{H_0(\zeta)} V(\zeta)\big) = 0.$$

Since $H_0(\zeta) = \zeta$, this is precisely (5.3). □

On $\Omega_0 \cup \Omega_h$, define the hyperbolic derivative of H_t and its first variation by

$$\Lambda_t(z) = \frac{(1 - |z|^2)|H_t'(z)|}{1 - |H_t(z)|^2}, \qquad q(z) = \operatorname{Re} V'(z) + \frac{2 \operatorname{Re}(\bar{z} V(z))}{1 - |z|^2}. \tag{5.4}$$

Here the derivatives are ordinary holomorphic derivatives, since H_t and V are holomorphic on the complementary domains.

Indeed, for every fixed $z \in \Omega_0 \cup \Omega_h$, Proposition 5.1 gives

$$H_t(z) = z + tV(z) + O_z(t^2), \qquad H_t'(z) = 1 + tV'(z) + O_z(t^2).$$

Consequently,

$$|H_t'(z)| = 1 + t \operatorname{Re} V'(z) + O_z(t^2),$$
$$1 - |H_t(z)|^2 = 1 - |z|^2 - 2t \operatorname{Re}(\bar{z} V(z)) + O_z(t^2).$$

Substitution into (5.4) yields $\Lambda_t(z) = 1 + tq(z) + O_z(t^2)$. The expansion is locally uniform in $z \in \Omega_0 \cup \Omega_h$.

5.2. **Nontriviality of the conformal variation.** The next proposition is independent of the triangular tessellation and the value of A_{AG}. It shows that the conformal variation cannot be infinitesimally trivial on both complementary domains.

Proposition 5.2. *Let F be holomorphic on $\mathbb{D}$ and have at least one critical point. Suppose that $u \in W^{1,\infty}_{\mathrm{loc}}(\mathbb{D})$ is continuous, takes the distinct constant values 0 and h on two domains Ω_0 and Ω_h, respectively, and satisfies $\partial_{\bar z} u = F' \partial_{\bar z} V$ for some continuous $V \in W^{1,p}_{\mathrm{loc}}(\mathbb{D})$, $p > 2$.*

Assume that V is holomorphic on both Ω_0 and Ω_h and extends holomorphically across a nonempty open arc of $\partial\mathbb{D}$ adjacent to each domain. Suppose also that $\operatorname{Re}(\bar\zeta V(\zeta)) = 0$ on these arcs. Then V''' cannot vanish identically on both Ω_0 and Ω_h.

Proof. Suppose, to the contrary, that V''' vanishes identically on both domains. Since Ω_0 and Ω_h are connected, there are quadratic polynomials K_0 and K_h such that $V = K_j$ on Ω_j, $j = 0, h$.

We first determine the form of these polynomials. Write $K_j(z) = A_j + B_j z + C_j z^2$. By hypothesis, $\operatorname{Re}(\bar\zeta K_j(\zeta)) = 0$ on a nonempty open arc of $\partial\mathbb{D}$. Since $|\zeta| = 1$, this becomes

$$\operatorname{Re}\bigl(A_j \bar\zeta + B_j + C_j \zeta\bigr) = 0.$$

The left-hand side is a trigonometric polynomial. As it vanishes on an open interval, it vanishes identically on $\partial\mathbb{D}$. Comparing Fourier coefficients gives $\operatorname{Re} B_j = 0$ and $C_j = -\overline{A_j}$. Thus

$$K_j(z) = \alpha_j + i\beta_j z - \overline{\alpha_j} z^2, \qquad \alpha_j \in \mathbb{C}, \quad \beta_j \in \mathbb{R}.$$

These are precisely the holomorphic vector fields generating one-parameter groups of disk automorphisms.

Set $W = V - K_0$ and $\Psi = u - F'W$. Since K_0 is holomorphic, $\partial_{\bar z} W = \partial_{\bar z} V$. The assumptions on u and V, together with the holomorphy of F', imply that $\Psi \in W^{1,p}_{\mathrm{loc}}(\mathbb{D})$ and is continuous. Moreover, the distributional product rule gives

$$\partial_{\bar z}\Psi = \partial_{\bar z} u - F' \partial_{\bar z} W = \partial_{\bar z} u - F' \partial_{\bar z} V = 0.$$

By Lemma 2.2, Ψ is holomorphic on $\mathbb{D}$.

On Ω_0, we have $u = 0$ and $W = V - K_0 = 0$. Hence Ψ vanishes on Ω_0, and the identity theorem gives $\Psi \equiv 0$ throughout $\mathbb{D}$.

On Ω_h, we have $u = h$ and $W = K_h - K_0$. It follows that

$$F'(K_h - K_0) = h$$

on Ω_h. Both sides are holomorphic on $\mathbb{D}$, so the identity theorem extends this equality to all of $\mathbb{D}$.

Let v be a critical point of F. Evaluating the identity at v gives $0 = F'(v)(K_h(v) - K_0(v)) = h$, contrary to $h \neq 0$. This contradiction proves the proposition. □

For the functions u and V defined in (4.3) and (5.2), respectively, we have

$$\partial_{\bar z} u = \frac{i}{2} \mathbf{1}_U \overline{F'} = F' \partial_{\bar z} V$$

almost everywhere on $\mathbb{D}$. The zeros of F' are isolated and do not affect this identity. Together with Proposition 5.1, this verifies the hypotheses of Proposition 5.2.

We also note that adding an infinitesimal disk automorphism to V does not change q. Indeed, if $K(z) = \alpha + i\beta z - \bar\alpha z^2$ with $\alpha \in \mathbb{C}$ and $\beta \in \mathbb{R}$, then

$$\operatorname{Re} K'(z) = -2 \operatorname{Re}(\bar\alpha z), \qquad \operatorname{Re}(\bar z K(z)) = (1 - |z|^2) \operatorname{Re}(\bar\alpha z).$$

It follows directly from (5.4) that $\operatorname{Re} K'(z) + 2\operatorname{Re}(\bar z K(z))/(1 - |z|^2) = 0$.

5.3. **Boundary behavior and triangle centers.** We first obtain a boundary expansion for q. Together with the bounded hyperbolic distance of points in $\mathbb{D}$ from $\mathcal{P}$, this will allow us to find a face center at which q is nonzero.

Lemma 5.3. *Let $J \subset \partial\mathbb{D}$ be a closed arc. Suppose that V is holomorphic in a neighborhood of J and satisfies (5.3) on J. Define q by (5.4). Then $\phi(\zeta) = \zeta^2 V'''(\zeta)$ is real-valued on J. Moreover, there exist constants $M > 0$ and $r_J \in (0,1)$ such that*

$$\left| q(r\zeta) - \frac{(1-r^2)^2}{12} \phi(\zeta) \right| \le M(1-r)^3, \qquad \zeta \in J, \quad r_J < r < 1. \tag{5.5}$$

Proof. Write $\zeta = e^{i\theta}$ and set $b(\theta) = \bar{\zeta}V(\zeta)$. The tangency condition (5.3) means that b is purely imaginary on J. Differentiating with respect to θ, we obtain

$$\begin{aligned} b' &= i(V' - b), \\ b'' &= -\zeta V'' + V' - b, \\ b''' + b' &= -i\zeta^2 V''', \end{aligned}$$

where the derivatives of V are evaluated at ζ. Since b and all its derivatives are purely imaginary, the last identity shows that $\zeta^2 V'''(\zeta) \in \mathbb{R}$. The second identity also gives $\operatorname{Re}(\zeta V''(\zeta)) = \operatorname{Re} V'(\zeta)$.

For $\zeta \in J$, put $a_\zeta(r) = \operatorname{Re}(\bar{\zeta}V(r\zeta))$. The preceding identities imply that

$$a_\zeta(1) = 0, \qquad a_\zeta''(1) = a_\zeta'(1), \qquad a_\zeta'''(1) = \phi(\zeta).$$

Furthermore, (5.4) gives

$$q(r\zeta) = a_\zeta'(r) + \frac{2r}{1-r^2} a_\zeta(r).$$

Set $x = 1 - r$, $A = a_\zeta'(1)$, and $B = \phi(\zeta)$. Taylor's formula yields

$$\begin{aligned} a_\zeta(1-x) &= -Ax + \frac{A}{2}x^2 - \frac{B}{6}x^3 + O_J(x^4), \\ a_\zeta'(1-x) &= A - Ax + \frac{B}{2}x^2 + O_J(x^3). \end{aligned}$$

Here the remainders are uniform for $\zeta \in J$, since V is holomorphic in a neighborhood of the compact arc J and its derivatives through order four are bounded there.

Using

$$\frac{2r}{1-r^2} = \frac{1}{x} - \frac{1}{2} - \frac{x}{4} + O(x^2),$$

we obtain

$$q(r\zeta) = \frac{B}{3}x^2 + O_J(x^3) = \frac{\phi(\zeta)}{3}(1-r)^2 + O_J((1-r)^3).$$

Finally,

$$\frac{(1-r^2)^2}{12} = \frac{(1-r)^2}{3} + O((1-r)^3).$$

Since ϕ is bounded on J, the last two estimates give (5.5). □

We now apply this expansion to the set $\mathcal{P}$ introduced in Proposition 3.2.

Proposition 5.4. *For the variation in Proposition 5.1, there exist $\epsilon \in \{-1, 1\}$ and infinitely many distinct points $p \in \mathcal{P} \cap (\Omega_0 \cup \Omega_h)$ such that $\epsilon q(p) > 0$.*

Proof. By Proposition 5.2, V''' cannot vanish identically on both complementary domains. Thus there is $j \in \{0, h\}$ such that $V''' \not\equiv 0$ on Ω_j.

Let I be the open arc of $\partial\mathbb{D}$ adjacent to Ω_j. By Proposition 5.1, V extends holomorphically across every closed subarc of I. The identity theorem implies that V''' cannot vanish identically on I. By Lemma 5.3, $\phi(\zeta) = \zeta^2 V'''(\zeta)$ is real-valued there. Consequently, we may choose a closed subarc $J \subset I$, a sign $\epsilon \in \{-1, 1\}$, and a constant $m > 0$ such that

$$\epsilon\phi(\zeta) \geq m, \qquad \zeta \in J.$$

Applying (5.5) and increasing r_J if necessary, we obtain

$$\epsilon q(r\zeta) > 0, \qquad \zeta \in J, \quad r_J < r < 1.$$

Fix ζ_* in the interior of J and choose a sequence $r_n \uparrow 1$. Set $z_n = r_n\zeta_*$. By Proposition 3.2, there are points $p_n \in \mathcal{P}$ such that $d_{\mathbb{D}}(z_n, p_n) \leq D_0$ for every n.

Put $\eta = \tanh(D_0/2) < 1$. By the formula for the Poincaré distance,

$$\left| \frac{p_n - z_n}{1 - \overline{z_n} p_n} \right| \leq \eta.$$

Since

$$|1 - \overline{z_n} p_n| \leq 1 - |z_n|^2 + |z_n||p_n - z_n|,$$

we obtain

$$|p_n - z_n| \leq \frac{\eta(1-|z_n|^2)}{1 - \eta|z_n|} \leq \frac{2\eta}{1-\eta}(1 - r_n).$$

In particular, $p_n \to \zeta_*$.

Since ζ_* lies in the interior of J, it follows that, for all sufficiently large n,

$$\frac{p_n}{|p_n|} \in J, \qquad |p_n| > r_J.$$

Moreover, $\zeta_* \notin \{a, b\}$. By (4.2), the boundary curves Γ_0 and Γ_h have no accumulation points on I. A sufficiently small neighborhood of ζ_* inside $\mathbb{D}$ therefore lies entirely in Ω_j. Hence $p_n \in \Omega_j$ for all sufficiently large n.

The sign estimate obtained above now gives $\epsilon q(p_n) > 0$ for all sufficiently large n. Finally, since $p_n \to \zeta_* \in \partial\mathbb{D}$, the sequence contains infinitely many distinct points. This proves the proposition. □

6. A bound for all univalent disks

We bound the Bloch radius of $\widetilde{f}_t$ using the triangular faces. The key observations are that the vertices remain critical under the deformation and that every point of a Euclidean triangle lies within its circumradius of some vertex.

Lemma 6.1. *For every real $|t| < 1/2$, the critical points of $\widetilde{f}_t$ are precisely the points $H_t(v)$ with $v \in \mathrm{Crit}(F)$, and each has local degree two. Moreover, G_t maps every closed lifted face homeomorphically onto a Euclidean triangle.*

Proof. Let Δ be a closed lifted face. Since the boundary curves of U consist of horizontal edges of the triangulation, the interior of Δ lies entirely in one of the regions U, Ω_0, or Ω_h. Accordingly, $G_t|_\Delta$ agrees with one of the maps

$$F + t\,\mathrm{Im}\,F, \qquad F, \qquad F + th.$$

These identities extend to the closed face by continuity. The first map is obtained by composing F with the horizontal shear $x + iy \mapsto x + ty + iy$, whose inverse is $x + iy \mapsto x - ty + iy$. Since F maps each closed lifted face homeomorphically onto a unit equilateral triangle, the same holds for G_t, with a possibly sheared image.

We next consider the critical points. Let $v \in \mathrm{Crit}(F)$. In a sufficiently small neighborhood of v, the definition (4.3) gives

$$|u(z) - u(v)| \le |F(z) - F(v)|.$$

Indeed, if v lies outside the boundary of U, the function u is locally constant. Otherwise, v lies on one of the boundary curves, where $u(v) = \mathrm{Im}\,F(v) \in \{0, h\}$. On each incident sector, u is either constant or equal to $\mathrm{Im}\,F$, so the inequality follows in either case.

By Lemma 3.1, F has local degree two at v. We may therefore choose a disk $D(v, \rho)$ such that $F(z) \neq F(v)$ for $0 < |z - v| \le \rho$. For $|t| < 1$, we have

$$\begin{aligned} |G_t(z) - G_t(v)| &\ge |F(z) - F(v)| - |t|\,|u(z) - u(v)| \\ &\ge (1 - |t|)|F(z) - F(v)| > 0, \qquad 0 < |z - v| \le \rho. \end{aligned} \tag{6.1}$$

Consider the homotopy

$$F(z) - F(v) + st\big(u(z) - u(v)\big), \qquad 0 \le s \le 1.$$

By (6.1), none of these maps vanishes on $\partial D(v, \rho)$. Their winding numbers around the origin are therefore equal. Since F has local degree two at v, the winding number of $G_t - G_t(v)$ is also two. Furthermore, (6.1) shows that v is the only preimage of $G_t(v)$ in $D(v, \rho)$. Thus the local topological degree of G_t at v is exactly two.

By Proposition 5.1, $\widetilde{f}_t = G_t \circ H_t^{-1}$ is holomorphic, and H_t is an orientation-preserving homeomorphism. Hence $\widetilde{f}_t$ has local degree two at $H_t(v)$, so $H_t(v)$ is a simple critical point of $\widetilde{f}_t$.

It remains to exclude other critical points. On the interior of each face, G_t is locally injective by its affine description. The same is true at an interior point of an edge not belonging to ∂U, since the adjacent faces have the same affine expression for G_t.

Now consider an interior point of an edge of ∂U. Since F' does not vanish there, we may use $w = F(z)$ as a local coordinate. The edge is mapped to a segment of the horizontal line $\mathrm{Im}\,w = c$, where $c = 0$ or h. On one side of the edge, G_t takes the form $w \mapsto w + t\,\mathrm{Im}\,w$, while on the other it takes the form $w \mapsto w + tc$. These maps agree on $\mathrm{Im}\,w = c$. Both preserve the imaginary coordinate, and for each fixed imaginary part their real components are translations. They therefore fit together to give a local homeomorphism across the edge.

Thus G_t is locally injective away from $\mathrm{Crit}(F)$. The same is true of $\widetilde{f}_t$ away from $H_t(\mathrm{Crit}(F))$. Since $\widetilde{f}_t$ is holomorphic, local injectivity is equivalent to having nonzero derivative. This proves the lemma. □

We shall also use the following elementary geometric observation, which holds for arbitrary Euclidean triangles, including obtuse ones.

Lemma 6.2. *Let Δ be a nondegenerate closed Euclidean triangle with circumradius R. Then every $w \in \Delta$ satisfies $|w - v| \le R$ for at least one vertex v of Δ.*

Proof. Let v_1, v_2, v_3 be the vertices of Δ, and let O be its circumcenter. Write $w = \sum_{j=1}^{3} \lambda_j v_j$, where $\lambda_j \ge 0$ and $\sum_{j=1}^{3} \lambda_j = 1$.

Since $|v_j - O| = R$ for each j, expanding the squares gives

$$\begin{aligned}\sum_{j=1}^{3} \lambda_j |w - v_j|^2 &= \sum_{j=1}^{3} \lambda_j |v_j - O|^2 - |w - O|^2 \\ &= R^2 - |w - O|^2 \le R^2.\end{aligned}$$

The left-hand side is a convex combination of the squared distances from w to the vertices. Hence at least one of these distances is at most R, as claimed. □

For the deformed triangles, set

$$R(t) = \frac{1}{\sqrt{3}} \sqrt{1 + \frac{3}{4}t^2 + \frac{9}{16}t^4}. \tag{6.2}$$

Proposition 6.3. *For every real $|t| < 1/2$, $b(\widetilde{f}_t) \le R(t)$.*

Proof. We first determine the circumradii of the images of the lifted faces.

On a face contained in U, the map G_t is obtained from F by the horizontal shear $x + iy \mapsto x + ty + iy$. Up to translation and a half-turn, the resulting triangle has vertices

$$0, \qquad 1, \qquad \frac{1}{2} + th + ih.$$

Its area is $h/2$, since the shear preserves area, and its squared side lengths are

$$1, \qquad 1 + ht + h^2t^2, \qquad 1 - ht + h^2t^2.$$

The circumradius of a triangle with side lengths a, b, c and area A is $abc/(4A)$. Thus the squared circumradius of a sheared face is

$$\begin{aligned}R_\Delta(t)^2 &= \frac{(1 + ht + h^2t^2)(1 - ht + h^2t^2)}{4h^2} \\ &= \frac{1 + h^2t^2 + h^4t^4}{4h^2} \\ &= \frac{1}{3}\left(1 + \frac{3}{4}t^2 + \frac{9}{16}t^4\right).\end{aligned}$$

Hence $R_\Delta(t) = R(t)$. Every face outside U is merely translated, so its circumradius remains $r_0 = 1/\sqrt{3} \le R(t)$. Consequently, the image of every closed lifted face is a Euclidean triangle of circumradius at most $R(t)$.

Suppose now that $\psi : D(w, r) \to \mathbb{D}$ is a holomorphic inverse branch of $\widetilde{f}_t$ for some $r > R(t)$. Set $z = \psi(w)$, so that $\widetilde{f}_t(z) = w$ and $\widetilde{f}_t'(z) \ne 0$.

Choose a closed lifted face Δ containing $H_t^{-1}(z)$. By Lemma 6.1, G_t maps Δ homeomorphically onto a Euclidean triangle containing w. Lemma 6.2 gives a vertex v of Δ such that $|w - G_t(v)| \le R(t) < r$. Since $G_t(\Delta)$ is convex, the segment joining w to $G_t(v)$ lies in $G_t(\Delta)$. Moreover, the whole segment is contained in $D(w, r)$.

Parametrize this segment by

$$\sigma(s) = (1 - s)w + sG_t(v), \qquad 0 \le s \le 1,$$

and define

$$\alpha(s) = H_t\big((G_t|_\Delta)^{-1}(\sigma(s))\big), \qquad 0 \le s \le 1.$$

The path α is continuous, and

$$\alpha(0) = z, \qquad \alpha(1) = H_t(v), \qquad \widetilde{f}_t \circ \alpha = \sigma.$$

For $0 \le s < 1$, the point $\alpha(s)$ is regular for $\widetilde{f}_t$. Indeed, the corresponding point of Δ is not a vertex: a segment joining a nonvertex point of a triangle to one of its vertices cannot pass through another vertex before reaching its endpoint. Lemma 6.1 therefore gives $\widetilde{f}_t'(\alpha(s)) \ne 0$.

We claim that $\alpha(s) = \psi(\sigma(s))$ for $0 \le s < 1$. This holds at $s = 0$. The set of parameters for which the two paths agree is closed by continuity. It is also relatively open in $[0, 1)$, since the local inverse theorem gives uniqueness of lifts near every regular point $\alpha(s)$. By connectedness, the two paths agree throughout $[0, 1)$.

Both paths extend continuously to $s = 1$, so

$$\psi(G_t(v)) = H_t(v).$$

But $H_t(v)$ is a critical point of $\widetilde{f}_t$ by Lemma 6.1. Differentiating $\widetilde{f}_t \circ \psi = \mathrm{id}$ at $G_t(v)$ gives

$$1 = \widetilde{f}_t'(H_t(v))\psi'(G_t(v)) = 0,$$

a contradiction.

Therefore no inverse branch exists on a disk of radius greater than $R(t)$, and hence $b(\widetilde{f}_t) \le R(t)$. □

The argument uses only the geometry of the face images and the local behavior of the map. More generally, suppose a holomorphic map admits a locally finite decomposition into closed triangular faces, maps each face homeomorphically onto a Euclidean triangle, has critical points at all vertices, and is regular along the open edges. Then its Bloch radius is bounded above by the supremum of the circumradii of the image triangles. No Delaunay condition or restriction on the centers of the univalent disks is needed.

7. Proof of the qualitative bound

We now prove Theorem 1.1. The essential point is to normalize the deformed map at a face center where the first variation of the hyperbolic derivative is nonzero. Choosing the sign of the deformation then gives a strict improvement over A_{AG}.

Proof. Normalization and comparison. By Proposition 5.4, there exists $p \in \mathcal{P} \cap (\Omega_0 \cup \Omega_h)$ such that $q(p) \neq 0$. Fix such a point and put $q_0 = q(p)$.

For $|t| < 1/2$, let $a_t = H_t(p)$ and $\varphi_a(z) = (z + a)/(1 + \bar{a}z)$. Define

$$f_{t,p}(z) = \frac{\widetilde{f}_t(\varphi_{a_t}(z)) - \widetilde{f}_t(a_t)}{(1 - |a_t|^2)\widetilde{f}_t'(a_t)}. \tag{7.1}$$

Since p lies in one of the complementary domains, u is constant in a neighborhood of p. Differentiating $\widetilde{f}_t \circ H_t = F + tu$ at p therefore gives

$$\widetilde{f}_t'(a_t)H_t'(p) = F'(p).$$

Here $F'(p) \neq 0$, since p is a face center, and $H_t'(p) \neq 0$, since H_t is conformal near p. Thus $f_{t,p}$ is well defined and holomorphic on $\mathbb{D}$. Moreover, $\varphi_{a_t}(0) = a_t$ and $\varphi_{a_t}'(0) = 1 - |a_t|^2$, so

$$f_{t,p}(0) = 0, \qquad f_{t,p}'(0) = 1.$$

Recall that precomposition with a disk automorphism leaves the Bloch radius unchanged, whereas multiplication by a nonzero constant scales it by the modulus of that constant. Consequently,

$$b(f_{t,p}) = \frac{b(\widetilde{f}_t)}{(1 - |a_t|^2)|\widetilde{f}_t'(a_t)|}.$$

On the other hand, the definition (5.4) of Λ_t, together with (3.1), gives

$$\begin{aligned}(1 - |a_t|^2)|\widetilde{f}_t'(a_t)| &= \frac{(1 - |H_t(p)|^2)|F'(p)|}{|H_t'(p)|} \\ &= \frac{(1 - |p|^2)|F'(p)|}{\Lambda_t(p)} \\ &= \frac{c_0}{\Lambda_t(p)}.\end{aligned}$$

Therefore Proposition 6.3 yields

$$b(f_{t,p}) \le \frac{R(t)}{c_0}\Lambda_t(p).$$

Since $r_0/c_0 = A_{\mathrm{AG}}$, the expression (6.2) for $R(t)$ gives

$$\frac{b(f_{t,p})}{A_{\mathrm{AG}}} \le \sqrt{1 + \frac{3}{4}t^2 + \frac{9}{16}t^4}\,\Lambda_t(p). \tag{7.2}$$

Choice of the deformation parameter. Since H_t is holomorphic near p and depends holomorphically on the parameter near $t = 0$, both $H_t(p)$ and $H_t'(p)$ are real-analytic functions of t. Moreover, $H_0 = \mathrm{id}$. It follows that $\Lambda_t(p)$ is real-analytic near $t = 0$, with

$$\Lambda_0(p) = 1, \qquad \frac{d}{dt}\Lambda_t(p)\bigg|_{t=0} = q_0.$$

Thus there exist $\varepsilon > 0$ and $C_p \geq 0$ such that

$$|\Lambda_t(p) - 1 - tq_0| \leq C_p t^2, \qquad \Lambda_t(p) \leq 2$$

whenever $|t| < \varepsilon$.

For $|t| \leq 1$, we also have

$$\sqrt{1 + \frac{3}{4}t^2 + \frac{9}{16}t^4} \leq 1 + t^2.$$

Combining these estimates with (7.2), we obtain

$$\frac{b(f_{t,p})}{A_{\mathrm{AG}}} \leq 1 + tq_0 + (C_p + 2)t^2 \tag{7.3}$$

for $|t| < \min\{\varepsilon, 1/2\}$.

Choose $\delta > 0$ such that

$$\delta < \min\left\{\varepsilon, \frac{1}{2}, \frac{|q_0|}{2(C_p + 2)}\right\}.$$

For $0 < s < \delta$, put $t = -\operatorname{sgn}(q_0)s$. Then $tq_0 = -|q_0|s$, and (7.3) gives

$$\begin{aligned}\frac{b(f_{t,p})}{A_{\mathrm{AG}}} &\leq 1 - |q_0|s + (C_p + 2)s^2 \\ &\leq 1 - \frac{|q_0|}{2}s.\end{aligned}$$

Thus, with $c = |q_0|/2 > 0$,

$$b(f_{t,p}) \leq A_{\mathrm{AG}}(1 - cs), \qquad 0 < s < \delta.$$

Convergence to the Ahlfors–Grunsky function. It remains to identify the limit of the normalized functions.

By Proposition 2.3(1), the maps H_t converge locally uniformly to the identity as $t \to 0$. Since they are quasiconformal self-homeomorphisms of $\mathbb{D}$, their inverses also converge locally uniformly to the identity.

The function u satisfies $0 \leq u \leq h$ on $\mathbb{D}$. Hence $G_t = F + tu$ converges locally uniformly to F. Together with the convergence of H_t^{-1}, this implies that $\widetilde{f}_t = G_t \circ H_t^{-1}$ converges locally uniformly to F. Since each $\widetilde{f}_t$ is holomorphic, the Cauchy integral formula also gives local uniform convergence of its derivatives.

In particular, $a_t = H_t(p) \to p$ and

$$\widetilde{f}_t(a_t) \longrightarrow F(p), \qquad \widetilde{f}_t'(a_t) \longrightarrow F'(p).$$

Furthermore, φ_{a_t} converges locally uniformly to φ_p. It follows from (7.1) that

$$f_{t,p} \longrightarrow g := \frac{F \circ \varphi_p - F(p)}{(1 - |p|^2)F'(p)}$$

locally uniformly on $\mathbb{D}$.

As established after Lemma 3.3, g is a normalized Ahlfors–Grunsky function and satisfies $b(g) = A_{\mathrm{AG}}$.

Finally, setting $f_s = f_{-\operatorname{sgn}(q_0)s,p}$ for $0 < s < \delta$ gives

$$f_s \longrightarrow g \quad \text{locally uniformly on } \mathbb{D}, \qquad b(f_s) \leq A_{\mathrm{AG}}(1 - cs).$$

This proves (1.2) and completes the proof of Theorem 1.1. □

Proposition 5.4 provides infinitely many possible normalization centers at which q has the same nonzero sign. The constants c and δ obtained above may depend on the chosen center; no uniform lower bound for c is claimed.

8. The explicit bound

We now consider a different deformation, obtained by shearing the first cylinder of the compact surface X. Unlike the deformation in Section 4, it acts on every lift of that cylinder. We derive a finite-parameter estimate for the resulting normalized functions and express its first-order term as an integral over the compact surface. The enclosure in Lemma 8.3 then yields the explicit bound.

8.1. **The deformation and its normalization.** For real t, replace the first copy of E in the construction of X by

$$E_t = \mathbb{C}/\big(2\mathbb{Z} + (\tau + th)\mathbb{Z}\big),$$

leaving the second copy unchanged, and glue the two tori crosswise along $[0,1]$ as before. Write (X_t, ω_t) for the resulting translation surface. The shear $A_t(x,y) = (x + ty, y)$ on the first torus and the identity on the second agree along the banks of the slit, since the slit lies on $y = 0$. They therefore define a homeomorphism $X \to X_t$. This is a deformation of the compact translation surface, rather than a separate deformation of its triangular faces.

Let p_X be the center of the upward face in the first torus with flat coordinate $w_* = 3/2 + ih/3$. In this section choose the universal covering $\pi : \mathbb{D} \to X$ so that $\pi(0) = p_X$, and choose the developing map F with $F(0) = w_*$. By Proposition 3.2, $|F'(0)| = c_0$. The threefold rotational symmetry of the conformal map between the centered hyperbolic and Euclidean triangles gives

$$F(z) = w_* + F'(0)z + O(z^4), \qquad \frac{F'(z)}{F'(0)} = 1 + O(z^3) \quad (z \to 0).$$

Indeed, the two centered triangles are invariant under rotations through $2\pi/3$, and uniqueness of the conformal triangle map gives the stated local equivariance. We shall use this expansion in the estimates below.

Let $\chi = \mathbf{1}_{\pi^{-1}(C)}$, where C is the first cylinder of X. On X consider the real one-form which equals $d\operatorname{Im} w$ on C and vanishes on its complement. It is closed in the distributional sense: its tangential trace is zero along each horizontal interface, and near a zero of ω its coefficients in a regular local coordinate are $O(|z - v|)$. In particular, there is no additional distributional term at the vertices. Its pullback to the simply connected disk thus has a continuous, locally Lipschitz primitive u_C, uniquely determined by $u_C(0) = 0$. In particular,

$$du_C = \chi \, d\operatorname{Im} F \quad \text{almost everywhere on } \mathbb{D}.$$

Set $G_t^C = F + tu_C$. The computation in Lemma 4.2, with χ in place of $\mathbf{1}_U$, gives a Beltrami coefficient of norm less than one. Let $K_t : \mathbb{D} \to \mathbb{D}$ be a quasiconformal solution of that equation, normalized by $K_t(0) = 0$, and define

$$\mathcal{F}_t = G_t^C \circ K_t^{-1}, \qquad c_t = |\mathcal{F}_t'(0)|, \qquad \widehat{f}_t = \frac{\mathcal{F}_t - \mathcal{F}_t(0)}{\mathcal{F}_t'(0)}. \tag{8.1}$$

By Lemma 2.4, $\mathcal{F}_t$ is holomorphic. As p_X is a regular point of F, the local degree of G_t^C at 0 is one. Hence $c_t > 0$, and $\widehat{f}_t(0) = 0$, $\widehat{f}_t'(0) = 1$.

Every lifted face is mapped by G_t^C onto either a translated unit equilateral triangle or its horizontal shear. The arguments proving Lemma 6.1 and Proposition 6.3 apply to this family as well: the face maps are homeomorphisms, the vertices retain local degree two, and the inverse-path argument excludes larger univalent disks. Using $R(t)$ from (6.2), we obtain

$$\log \frac{b(\widehat{f}_t)}{A_{\mathrm{AG}}} \le \log \frac{c_0}{c_t} + \frac{1}{2} \log \left(1 + \frac{3}{4}t^2 + \frac{9}{16}t^4\right). \tag{8.2}$$

Here we used $c_0 = 1/(\sqrt{3}\, A_{\mathrm{AG}})$. The bound holds for every inverse branch; no assumption on the position of its center is needed.

8.2. **A finite-parameter comparison.** For $F'(z) \neq 0$, let $Q(z) \in \mathrm{SO}(2)$ be the real rotation corresponding to $F'(z)/|F'(z)|$, and write $Q_0 = Q(0)$. Its value at a critical point may be chosen arbitrarily. Consider the symmetric matrix field

$$B_t = I + tC_1 + t^2 C_2,$$
$$C_1 = \chi Q^{\mathsf{T}} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} Q, \qquad C_2 = \chi Q^{\mathsf{T}} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} Q.$$

Indeed, on $\{\chi = 1\}$ the differential of G_t^C is the composition of the conformal differential of F with the shear matrix $A_t = \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$; on $\{\chi = 0\}$ it is conformal. Consequently,

$$B_t = (\det DG_t^C)(DG_t^C)^{-1}(DG_t^C)^{-\mathsf{T}} \quad \text{almost everywhere.}$$

This matrix is unchanged by holomorphic postcomposition, so the same formula holds with K_t in place of G_t^C. Since $K_t(0) = 0$ and $K_t(\mathbb{D}) = \mathbb{D}$, the function $g_t = -\log|K_t|$ is the Green function for the corresponding divergence-form operator:

$$-\operatorname{div}(B_t \nabla g_t) = 2\pi\delta_0 \quad \text{in } \mathbb{D}, \qquad g_t|_{\partial\mathbb{D}} = 0.$$

Both the equation and the boundary condition are understood in the standard weak and trace senses, respectively.

To compare g_t with an explicit Green function having the same leading singularity, put $a_0 = 1/100$, $b_0 = 1/50$, and define

$$\kappa(r) = \begin{cases} 1, & 0 \le r \le a_0, \\ 2 - r/a_0, & a_0 < r < b_0, \\ 0, & r \ge b_0. \end{cases}$$

Regard complex numbers as real column vectors and set

$$S = Q_0^{\mathsf{T}} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} Q_0, \qquad \phi_t(z) = z + t\kappa(|z|)Sz.$$

The bounds $|\kappa(|z|)Sz| \le |z|$ and $\|D(\kappa(|z|)Sz)\| \le 3$ show that ϕ_t is bi-Lipschitz for $|t| \le 1/10$. Since it is the identity near $\partial\mathbb{D}$, it maps $\mathbb{D}$ onto itself. Define

$$\begin{aligned} \widehat{B}_t &= (\det D\phi_t)(D\phi_t)^{-1}(D\phi_t)^{-\mathsf{T}}, \\ \widehat{g}_t &= -\log|\phi_t|, \\ I(t) &= \frac{1}{2\pi}\int_{\mathbb{D}} \nabla\widehat{g}_t^{\mathsf{T}}(B_t - \widehat{B}_t)\nabla\widehat{g}_t \, dA. \end{aligned}$$

The integral is well defined: the two matrix fields agree to sufficiently high order at the origin, where $\nabla\widehat{g}_t = O(|z|^{-1})$.

Proposition 8.1. *For $|t| \le 1/10$, let $v_t = g_t - \widehat{g}_t$. Then $v_t \in W_0^{1,2}(\mathbb{D})$ and*

$$\log\frac{c_t}{c_0} = -I(t) + \frac{1}{2\pi}\int_{\mathbb{D}} \nabla v_t^{\mathsf{T}} B_t \nabla v_t \, dA. \tag{8.3}$$

Proof. The singularity at the origin. Near 0, the coefficient of the Beltrami equation is smooth and G_t^C is the horizontal shear of F. Since $\mathcal{F}_t$ is locally univalent, comparison of the linear terms in $\mathcal{F}_t \circ K_t = G_t^C$ gives

$$|K_t(z)| = \frac{c_0}{c_t}|A_t Q_0 z| + O(|z|^2).$$

For $|z| < a_0$, we also have $|\phi_t(z)| = |A_t Q_0 z|$. The leading logarithmic singularities therefore cancel, and v_t extends continuously to 0 with $v_t(0) = \log(c_t/c_0)$. Local smoothness gives finite Dirichlet energy near 0. Away from the origin, the quasiconformal change-of-variables formula gives finite energy; both functions have zero boundary trace. Thus $v_t \in W_0^{1,2}(\mathbb{D})$.

The energy identity. Put $D_t = B_t - \widehat{B}_t$ and $\mathcal{R}_t = \operatorname{div}(D_t\nabla\widehat{g}_t)$. Subtracting the two Green equations gives

$$-\operatorname{div}(B_t\nabla v_t) = \mathcal{R}_t.$$

The expansion $F'(z)/F'(0) = 1 + O(z^3)$ implies that $D_t = O((|t| + t^2)|z|^3)$ near 0, since the reference matrix agrees there with the constant-rotation model. Consequently $D_t\nabla\widehat{g}_t = O(|z|^2)$ near 0; away from 0 this vector field is square integrable. In particular, all the following pairings are justified. Testing with v_t yields

$$\langle \mathcal{R}_t, v_t\rangle = \int_{\mathbb{D}} \nabla v_t^{\mathsf{T}} B_t \nabla v_t \, dA.$$

For a second test, set $\psi_\varepsilon = \min\{g_t, -\log\varepsilon\}$ and $\widetilde{v}_t = v_t \circ K_t^{-1}$. The change of variables $w = K_t(z)$ and integration in polar coordinates give

$$\begin{aligned} \int_{\mathbb{D}} \nabla v_t^{\mathsf{T}} B_t \nabla\psi_\varepsilon \, dA &= \int_{\varepsilon<|w|<1} \nabla\widetilde{v}_t \cdot \nabla(-\log|w|)\, dA(w) \\ &= \int_0^{2\pi} \widetilde{v}_t(\varepsilon e^{i\theta})\, d\theta \longrightarrow 2\pi v_t(0). \end{aligned}$$

The weak equation gives, for the same limit,

$$2\pi v_t(0) = -\int_{\mathbb{D}} D_t\nabla\widehat{g}_t \cdot \nabla g_t \, dA.$$

The bound near 0 justifies passage to the limit there, while Cauchy–Schwarz applies on its complement. Finally, use $g_t = \widehat{g}_t + v_t$ and $\int_{\mathbb{D}} \nabla v_t \cdot D_t\nabla\widehat{g}_t \, dA = -\int_{\mathbb{D}} \nabla v_t^{\mathsf{T}} B_t \nabla v_t \, dA$. The definition of $I(t)$ then gives (8.3). □

The first-order term in $I(t)$ is the real number

$$q_C = \frac{1}{2\pi} \operatorname{Im} \operatorname{pv} \int_{\mathbb{D}} \frac{\chi(z)\overline{F'(z)}}{F'(z)z^2}\, dA(z). \tag{8.4}$$

The principal value is taken by deleting $\{|z| < \varepsilon\}$ and letting $\varepsilon \downarrow 0$. At the origin the leading term of the integrand has zero angular mean, and the expansion of F' above ensures convergence of the remaining contribution.

Proposition 8.2. *For $0 < |t| \le 10^{-4}$,*

$$|I(t) - tq_C| < 9t^2.$$

Proof. The disk around the base point. The centered hyperbolic triangle contains $\{|z| \le 1/5\}$. Indeed, in the disk model its sides lie on circles whose distance from the origin is $C - \sqrt{C^2-1} > 1/5$, where $C^2 = (5+3\sqrt{3})/2$. Since its flat image is a unit equilateral triangle centered at w_*, $|F(z) - w_*| \le r_0$ on this disk. Cauchy's estimate gives

$$\left|\frac{F'(z)}{F'(0)}\right| \le \frac{20 r_0}{c_0} = 20A_{\mathrm{AG}} < 12, \qquad |z| \le \frac{1}{10}.$$

Only powers divisible by three occur in the Taylor series of $F'/F'(0)$. Another application of Cauchy's estimate therefore gives, for $r = |z| \le b_0$,

$$\left|\frac{F'(z)}{F'(0)} - 1\right| \le \frac{12000 r^3}{1 - 1000 r^3} < 12500 r^3.$$

This disk lies in $\{\chi = 1\}$. Let C_j^0 denote the matrices obtained from C_j by setting $\chi = 1$ and $Q = Q_0$. The preceding bound implies $\|C_j - C_j^0\| < 50000 r^3$ for $j = 1, 2$.

Put $p_t = r\nabla \widehat{g}_t$. Direct differentiation of $\widehat{g}_t$ gives, for $|t| \le 1/10$,

$$|p_t| \le \frac{9}{5}, \qquad |p_t - p_0| \le 8|t|.$$

For example, the denominator in the formula for p_t is bounded below by $(1-|t|)^2$, while the numerator and its difference from the value at $t = 0$ are bounded by $(1+3|t|)(1+|t|)$ and $5|t| + 4t^2$, respectively. Hence, for any real symmetric matrix M,

$$|p_t^{\mathsf{T}} M p_t - p_0^{\mathsf{T}} M p_0| \le 23 \|M\| |t|.$$

It follows that the contribution of the variable rotation to $I(t) - tq_C$, over $r < b_0$, has modulus at most

$$27 \cdot 50000 \int_0^{b_0} r^2\, dr\, t^2 = \frac{18}{5} t^2.$$

The transition annulus. Rotate coordinates so that $S = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Write $c = \cos\theta$, $s = \sin\theta$, $d = r\kappa'(r)$, $\alpha = dcs$, and $\beta = \kappa + ds^2$. Differentiating ϕ_t yields

$$\begin{aligned} L := D(\kappa S z) &= \begin{pmatrix} \alpha & \beta \\ 0 & 0 \end{pmatrix}, \\ \widehat{B}_t &= \begin{pmatrix} \dfrac{1 + t^2\beta^2}{1 + t\alpha} & -t\beta \\ -t\beta & 1 + t\alpha \end{pmatrix}, \\ p_t &= -\frac{(I + tL)^{\mathsf{T}}\big((c,s)^{\mathsf{T}} + t\kappa(s,0)^{\mathsf{T}}\big)}{1 + 2t\kappa cs + t^2\kappa^2 s^2}. \end{aligned} \tag{8.5}$$

Consider

$$\mathcal{J}(t; r, \theta) = p_t^{\mathsf{T}}\big(I + tC_1^0 + t^2 C_2^0 - \widehat{B}_t\big) p_t.$$

Its first-order coefficient has zero angular mean, and its second-order coefficient has angular mean

$$\frac{d^2 + 6d\kappa - 4d + 4\kappa^2 - 8\kappa + 4}{8}.$$

On $a_0 < r < b_0$, substituting $\kappa = 2 - r/a_0$ and $d = -r/a_0$ and integrating against dr/r gives $1/16 + (\log 2)/2$.

For the remaining terms, regard t as complex. On $|t| = 1/10$ the denominators in (8.5) have moduli at least $9/10$ and $89/100$, and $\|L\| \le 3$. In particular,

$$|\mathcal{J}(t; r, \theta)| \le \left(\frac{143}{89}\right)^2 \left(\frac{111}{100} + \frac{169}{90}\right) < 8.$$

Cauchy's estimate bounds the terms of degree at least three by $8000|t|^3/(1-10|t|) < 8010|t|^3$ for $|t| \le 10^{-4}$. Their total contribution, after radial integration, is at most $8010(\log 2)|t|^3$.

The exterior region. For $r \ge b_0$, the reference map is the identity. The term quadratic in t is nonnegative and at most $t^2 \log 50$. For $r < a_0$, the reference matrix agrees with the constant-rotation model. The first-order reference contribution has zero angular mean; the remaining first-order term is precisely q_C in (8.4). Combining the estimates gives

$$|I(t) - tq_C| \le \left(\frac{18}{5} + \frac{1}{16} + \log 50 + \left(\frac{1}{2} + \frac{801}{1000}\right)\log 2\right) t^2 < 9t^2.$$

The coefficient in parentheses is less than 8.477. □

The energy term in (8.3) is nonnegative. Combining that identity with (8.2) and Proposition 8.2, and using $\frac{1}{2}\log(1 + 3t^2/4 + 9t^4/16) \le t^2$ for $|t| \le 1$, we obtain

$$\log \frac{b(\widehat{f}_t)}{A_{\mathrm{AG}}} \le tq_C + 10t^2, \qquad 0 < |t| \le 10^{-4}. \tag{8.6}$$

8.3. **Reduction to a finite integral.** We next express q_C using differential forms on the compact surface. This eliminates the infinitely many lifts appearing in (8.4) and leads to finite integrals that can, in principle, be enclosed by interval arithmetic.

Let Γ be the deck group of π. Consider the Poincaré series

$$\mathcal{Q}(z)\,dz^2 = \sum_{\gamma\in\Gamma} \left(\frac{\gamma'(z)}{\gamma(z)}\right)^2 dz^2. \tag{8.7}$$

Away from the Γ-orbit of 0, the series converges locally uniformly. In fact, writing $a = \gamma^{-1}(0)$, a summand on a fixed compact set is $O((1-|a|^2)^2) = O(e^{-2d_{\mathbb{D}}(0,a)})$, whereas the number of orbit points at hyperbolic distance at most R is $O(e^R)$. Thus (8.7) defines a Γ-invariant meromorphic quadratic differential on $\mathbb{D}$ and descends to X. Its only pole on X is a double pole at p_X. In the flat coordinate w, its principal part is $dw^2/(w-w_*)^2$, with no simple-pole term. The latter assertion follows from $F(z) = w_* + F'(0)z + O(z^4)$. Throughout this subsection, principal values at the poles are taken by deleting centered disks in the local flat coordinate. This agrees with the convention in (8.4): the expansion $w - w_* = F'(0)z + O(z^4)$ shows that changing between the two cutoffs contributes $o(1)$.

Let P_1 be the horizontal period of $\mathcal{Q}/\omega$ on the first cylinder, with orientation in the positive real direction. The period is independent of the height of the cycle, since the pole of $\mathcal{Q}/\omega$ has zero residue. Unfolding the sum (8.7) in (8.4) gives

$$q_C = \frac{h}{2\pi}\operatorname{Im} P_1. \tag{8.8}$$

One must account for the centered principal value at p_X. In the coordinate $w - w_* = x + iy$, integration of $(x+iy)^{-2}$ over the chord $-\sqrt{\varepsilon^2 - y^2} < x < \sqrt{\varepsilon^2 - y^2}$ gives

$$-\frac{2\sqrt{\varepsilon^2 - y^2}}{\varepsilon^2}.$$

Integrating this expression over $0 < |y| < \varepsilon$ gives $-\pi$. Here the integration is performed first in x and then in y. The centered two-dimensional principal value of $(x+iy)^{-2}$ over the disk is zero. Consequently, deleting the disk after taking the full horizontal period contributes the real correction $+\pi$: the complex area integral is $hP_1 + \pi$. Since only its imaginary part occurs in (8.8), this correction does not affect q_C. Away from the pole the unfolding is absolutely convergent; the flat density near a branch point has at worst an integrable $O(|w - w_v|^{-1})$ singularity.

Let ι be the involution of X exchanging its two sheets over E. The anti-invariant meromorphic one-form

$$\mathcal{A}^- := \frac{\mathcal{Q} - \iota^*\mathcal{Q}}{2\omega}$$

is regular at the zeros of ω. Indeed, in a local coordinate with $\iota(\xi) = -\xi$ and $\omega = 2\xi\,d\xi$, the coefficient of the numerator is odd in ξ and therefore divisible by ξ. The only poles of $\mathcal{A}^-$ occur at p_X and ιp_X, with opposite principal parts $\frac{1}{2}\,dw/(w-w_*)^2$ and $-\frac{1}{2}\,dw/(w-w_*)^2$, respectively.

The anti-invariant space of holomorphic one-forms on X is one-dimensional. Let η generate it, and choose an anti-invariant meromorphic one-form $\mathcal{A}_0$ with these principal parts. Then $\mathcal{A}^- = \mathcal{A}_0 + c\eta$ for some $c \in \mathbb{C}$.

To determine c, write the hyperbolic metric of X as $\rho_X(z)|dz|$ and set $\rho_{\mathbb{D}}(z) = 2/(1-|z|^2)$. The Poincaré series is orthogonal, in the principal-value sense, to every holomorphic quadratic differential φ on X:

$$\operatorname{pv}\int_X \frac{\mathcal{Q}\,\overline{\varphi}}{\rho_X^2}\,dA = 0.$$

Indeed, unfolding reduces the integral to

$$\operatorname{pv}\int_{\mathbb{D}} \frac{\overline{\varphi(z)}}{z^2\rho_{\mathbb{D}}(z)^2}\,dA(z),$$

whose angular integral vanishes on every annulus centered at 0. The integral is absolutely convergent near $\partial\mathbb{D}$, since $|\varphi|/\rho_{\mathbb{D}}^2$ is bounded. The same orthogonality holds for $\iota^*\mathcal{Q}$.

Put $m = |\omega|_X^2$ and define the weighted pairing

$$\langle \alpha, \beta\rangle_m = \frac{i}{2}\operatorname{pv}\int_X m\,\alpha\wedge\overline{\beta}.$$

Applying orthogonality with the holomorphic quadratic differential $\varphi = \omega\eta$ gives

$$c = -\frac{\langle \mathcal{A}_0, \eta\rangle_m}{\langle \eta, \eta\rangle_m}.$$

The denominator is positive. In addition, the sum of the imaginary horizontal periods of $\mathcal{Q}/\omega$ on the two cylinders vanishes: replacing χ by 1 in (8.4) gives zero, as is seen from the threefold symmetry $F'(e^{2\pi i/3}z) = F'(z)$. Thus the imaginary period of $\mathcal{A}^-$ on the first cylinder equals $\operatorname{Im} P_1$.

We make the projection explicit on an elliptic model of X. All Weierstrass functions below have period lattice L_0. Set $a_* = 1/2$ and

$$f_E(w) = \wp(w - a_*) - \wp(a_*).$$

The compactification of

$$v^2 = f_E(w)$$

is a double cover of E, branched at 0 and 1. It gives the same marked cover as the slit construction. One way to verify the marking is to deform the lattice through $\tau_s = s/2 + ih$, $0 \le s \le 1$, and transport the cycles

$$a_s(x) = x + \tau_s/2 \quad (0 \le x \le 2), \qquad b_s(r) = 3/2 + r\tau_s \quad (0 \le r \le 1).$$

These cycles avoid 0, 1, and a_* modulo the corresponding period lattice. At $s = 0$, the function $\wp_s(w - a_*) - \wp_s(a_*)$ is negative real on both cycles, so its square root has trivial monodromy there. This monodromy remains unchanged during the deformation, while a small loop around either simple zero reverses the sign. These are precisely the monodromies of the slit double cover.

On this model $\omega = dw$ and $\eta = dw/v$. Choose the value $v_* = v(w_*)$ on the first sheet and set $v_*' = \wp'(w_* - a_*)/(2v_*)$. Define

$$R_*(w) = \frac{1}{2}\left[v_*\wp(w - w_*) + v_*'\big(\zeta_{L_0}(w - w_*) - \zeta_{L_0}(w - a_*)\big)\right]. \tag{8.9}$$

The differential $\mathcal{A}_0 = R_*(w)\,dw/v(w)$ has the required principal parts. At $w = w_*$, the simple-pole terms from $1/v$ and the zeta function cancel; on the other sheet both principal parts change sign. At $w = a_*$, the apparent pole of R_* is canceled by the zero of dw/v.

The hyperbolic metric, and hence m, is invariant under ι and may be regarded as a function of $w \in E$. Each weighted pairing is twice the corresponding integral over E, so the factor of two cancels in their ratio. Thus, writing $D_m = \langle \eta, \eta\rangle_m/2$ and $M_* = -c$, we obtain the finite formulas

$$D_m = \int_E \frac{m(w)}{|f_E(w)|}\,dA > 0, \qquad M_* = \frac{\operatorname{pv}\displaystyle\int_E \frac{m(w)R_*(w)}{|f_E(w)|}\,dA}{D_m},$$
$$q_C = \frac{h}{2\pi}\operatorname{Im}\int_{a_1}\frac{R_*(w) - M_*}{v(w)}\,dw. \tag{8.10}$$

Here a_1 is a positively oriented horizontal cycle in the first sheet, and v is continued along that cycle. The expression is unaffected by reversing the initial choice of square root. In the flat coordinate, $m(w)$ is the inverse square of the hyperbolic density.

8.4. **The numerical enclosure.** We describe the evaluation of (8.10) used to obtain Lemma 8.3. The implementation and computational records are available in the repository (outer, inner, and period logs).

Let ${}_2F_1$ be the Gauss hypergeometric function, and set

$$W(\xi) = c\xi\, {}_2F_1(1/3, 2/3; 4/3; \xi^3), \qquad c = \frac{\Gamma(2/3)}{\sqrt{3}\,\Gamma(4/3)\Gamma(1/3)},$$
$$Z(\xi) = k\xi\, \frac{{}_2F_1(3/4, 5/12; 4/3; \xi^3)}{{}_2F_1(5/12, 1/12; 2/3; \xi^3)}, \quad k = \frac{\Gamma(2/3)\Gamma(11/12)}{\sqrt{1+\sqrt{3}}\,\Gamma(4/3)\Gamma(1/4)}.$$

These are the flat and hyperbolic triangle coordinates, respectively; compare the Schwarz triangle formula in Ahlfors–Grunsky [3, Lemma, p. 672]. For the upward faces of E take $w = j + 1/2 + ih/3 + e^{7\pi i/6} W(\xi)$, $j = 0, 1$; the downward faces are obtained by $w \mapsto 1 - w$ modulo L_0. Since the hyperbolic density in the flat coordinate is $2|Z'|/((1-|Z|^2)|W'|)$, the measure appearing in (8.10) becomes

$$m(w)\, dA(w) = \frac{(1-|Z(\xi)|^2)^2 |W'(\xi)|^4}{4|Z'(\xi)|^2}\, dA(\xi). \tag{8.11}$$

The choice of phase agrees with the indicated vertices and the base point w_*.

For the inner disks $|\xi| \le r_i = 3/4$, put $b(\xi) = (dw/d\xi)^2/(v(w)Z'(\xi))$. Expanding bZ^j and $R_* bZ^j$, $j = 0, 1, 2$, reduces the relevant integrals to orthogonal monomial moments. At the double pole, the principal part is subtracted before the series is integrated; terms with negative powers have zero angular mean against the conjugate holomorphic series. Suppose that two regular series have modulus bounds M_1, M_2 on $|\xi| = r_c = 7/8$. Their product tail beyond degree $N = 160$, including the factor $1/4$ in (8.11), is bounded by

$$\frac{\pi r_i^2 M_1 M_2}{4(N+2)} \frac{(r_i/r_c)^{2N+2}}{1-(r_i/r_c)^2}. \tag{8.12}$$

A certificate must supply rigorous bounds M_1, M_2 on the entire circle. The downward faces may be paired using the involution $w \mapsto 1 - w$, replacing $R_*(w)$ by $R_*(w) + R_*(1-w)$.

For the remaining annular sectors, one may use the parametrizations

$$\xi = e^{2\pi i j/3} \left(1 - \frac{1}{4} a s^6\right) \exp\left(\pm \frac{\pi i}{3} b s^6\right), \qquad j = 0, 1, 2,$$

where $(a, b) = (1, t)$ or $(t, 1)$ and $(s, t) \in [0, 1]^2$. The absolute area Jacobian is $\frac{\pi}{2}(1 - \frac{1}{4}as^6)s^{11}$. The connection formulas at $\xi^3 = 1$ put the fractional corner powers in the form $1 - \xi^3 = s^6 D$ and the flat displacement in the form $s^2 L$. The resulting integrands must be bounded, together with their required derivatives, on every accepted rectangle. For a rectangle of side lengths ℓ_1, ℓ_2, the error bound for the tensor-product four-point Gauss rule is

$$\frac{\ell_1 \ell_2}{44100} \left(\ell_1^8 \sup \left| \frac{\partial_1^8 I}{8!} \right| + \ell_2^8 \sup \left| \frac{\partial_2^8 I}{8!} \right| \right). \tag{8.13}$$

The derivative bounds in (8.13) may be obtained by interval automatic differentiation. Equation (8.12) controls the inner-disk tail.

The periods may be evaluated by a two-point Gauss rule, whose error on a segment of length ℓ is at most $\ell^5 \sup |I^{(4)}/4!|/180$. For analytic continuation of the square root, it suffices to enclose the ratio of f_E to its value at the beginning of each segment in the right half-plane. Paths at heights $h/2$ and $7h/10$ provide an independent consistency check.

Lemma 8.3. *The expression for q_C in* (8.10)*, with* (8.9) *and* (8.11)*, satisfies*

$$-\frac{370}{10^6} < q_C < -\frac{337}{10^6}. \tag{8.14}$$

The numerical data reported for this computation are

$$D_m = 0.13566064254886\ldots \pm 3.76 \cdot 10^{-7},$$
$$D_m M_* = (0.09734640938705\ldots \pm 1.52 \cdot 10^{-5}) + i(0.13409159372508\ldots \pm 1.52 \cdot 10^{-5}),$$

with both evaluations of q_C in $[-0.00036984, -0.00033743]$. These values include the factor $h/(2\pi)$ in (8.10). The resulting interval is contained in $(-370/10^6, -337/10^6)$, which proves (8.14).

8.5. **The explicit bound.** Now we establish the following result.

Proposition 8.4. *Let $t_* = 337/20000000$. Then the compact-cylinder family in* (8.1) *satisfies*

$$\mathcal{B} \le b(\widehat{f}_{t_*}) < A_{\mathrm{AG}} \exp\left(-\frac{113569}{40000000000000}\right) < 0.471861652113 < A_{\mathrm{AG}} - \frac{13}{10^{10}}. \tag{8.15}$$

Proof. Set $a = 337/10^6$ and $t_* = a/20$, so that $0 < t_* < 10^{-4}$. The following identity is exact:

$$-at_* + 10t_*^2 = -\frac{a^2}{40} = -d_*, \qquad d_* := \frac{113569}{40000000000000}.$$

By (8.14) and (8.6),

$$\log \frac{b(\widehat{f}_{t_*})}{A_{\mathrm{AG}}} < t_* \left(-\frac{337}{10^6}\right) + 10t_*^2 = -d_*.$$

This proves the first strict inequality in (8.15).

The remaining scalar comparisons follow, for example, from the following high-precision bounds for the Gamma function and exponential:

$$0.47186165345268178487 < A_{\mathrm{AG}} < 0.47186165345268178488,$$

$$0.47186165211296038375 < A_{\mathrm{AG}} e^{-d_*} < 0.47186165211296038376.$$

These scalar bounds are independent of the two-dimensional integrals in (8.10). They imply

$$A_{\mathrm{AG}} e^{-d_*} < \frac{471861652113}{10^{12}} < A_{\mathrm{AG}} - \frac{13}{10^{10}}.$$

The definition of $\mathcal{B}$ now gives (8.15). □

Proposition 8.4 uses the compact-surface deformation: the coefficient q_C and the remainder estimate in Proposition 8.2 refer to that family. Theorem 1.1 uses the single lifted cylinder from Section 4.

Proof of Theorem 1.3. This is an immediate corollary of Proposition 8.4. □

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